Optimising Binary HVAC Switching in Saudi Residential Buildings with Model Predictive Control and Rule-Constrained PPO

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Abstract

On/off cooling of Saudi residential split units is posed as a constrained optimisation with hard comfort constraints, an energy cost and a charge per on-command, and simulated for Riyadh, Jeddah and Taif on held-out weeks against an assumed 1 °C-deadband thermostat shifted to each controller's mean temperature. Tuning its two switch points widens the deadband: it avoids about 22–24% of on-commands with electricity almost unchanged and saves 12.4–13.6% of cost at a reference charge above the range a wear estimate supports; against a wider-deadband comparator or on a real thermostat's 0.1 °C grid it saves much less. The saving prices avoided on-commands, not a lower bill. A 24 h model predictive controller (MPC) adds about 4–6 points with a perfect in-horizon forecast; with persistence its margin is not resolved. Both depend strongly on the assumed thermal capacity, whose reference value probably favours the tuned thermostat, and on the actuation step: the tuned thermostat saves about 25% at 5 min and nothing resolved at 30 min. Proximal policy optimisation inside hard rules reproduces the rules' hold policy in Riyadh and Jeddah; in Taif it falls below it by an unresolved margin. A clairvoyant schedule and a provable bound place the best attainable saving at about 17–21% at that charge. In two- and four-room dwellings the tuned thermostat's saving is not resolved from the single-zone value. In this model, electricity at matched comfort follows the comfort target and the outdoor air, not switching control.

Keywords: Binary HVAC switching, model predictive control, constrained reinforcement learning, Saudi Vision 2030, Green Riyadh, energy efficiency

1. Introduction

Buildings consume approximately 40% of global final energy, with heating, ventilation and air-conditioning (HVAC) systems accounting for roughly half that figure [1]. In Saudi Arabia the share is larger: residential buildings alone consume ≈50% of national electricity, and air conditioning represents up to 70% of residential demand [2, 3]. Residential air conditioning is therefore central to the efficiency targets of Saudi Vision 2030 [4], whose Saudi Green Initiative commits to cutting emissions by more than 278 Mt a year by 2030 and to net zero by 2060 [5]; in the capital, the Green Riyadh programme of more than 7.5 million trees aims to lower temperatures in the city by 1.5–2 °C [6]. Section 10.3 asks how much of that switching control can deliver, against the comfort target and a cooler city.

1.1. Motivation

The prevalent residential control strategy is the automatic thermostat: it switches on when a room exceeds an upper setpoint and off when it falls to a lower one, with no tariff awareness and no inter-zone coordination. It cycles on a fixed deadband, and every on-command adds wear that it does not price. Model predictive control (MPC) plans against a model and a forecast [7]; reinforcement learning (RL) learns a policy from interaction without an explicit model [8, 9, 10], yet pure RL agents can violate comfort constraints during exploration, which hybrid approaches prevent by combining hard safety rules with RL [11]. The closest prior work is the present author's physics-informed proximal policy optimisation (PPO) framework for

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Saudi residential split units under the same tariff [12], which Section 2.4 positions against the present study.

1.2. Contributions

Six contributions are made. (C1) A constrained binary formulation of residential on/off cooling, over single- and multi-zone topologies, with a hard occupied ceiling and a hard cold rule (a cooling step may never end below the band; free-float is unrestricted), a charge per supervisory on-command and a tariff-weighted energy cost, and three optimality conditions under a limit-average criterion. (C2) A connection to Simple Linear Hybrid Automata [13] which shows that, under constant disturbance and at the mean of its own maximal-width cycle, the hysteresis induced by the hard rules is optimal among policies that act only where the rules leave the action free (Corollary 1), and which names the features that remove that optimality, a diurnal disturbance first among them. (C3) One selection and evaluation protocol applied identically to every admitted arm: parameters chosen on tuning weeks under a single admissibility rule with a single fallback (and, for MPC alone, an occupied-ceiling back-off), and each held-out week compared with a thermostat shifted to the arm's own clipped mean occupied temperature, with pooled savings and exact paired statistics in pre-declared families. (C4) A controlled comparison of a tuned two-parameter thermostat, rule-constrained PPO, a one-step model shield and MPC at horizons from 15 min to 48 h, with perfect and persistence forecasts and across thermal capacities. The tuned thermostat's gain is a wider effective deadband. MPC's margin over it is resolved only with a perfect in-horizon forecast, shrinks with thermal capacity and changes with the actuation step. An exact one-step shield in place of the rules' fixed margins matches the tuned thermostat, and PPO inside the rules reproduces the rules' hold policy in Riyadh and Jeddah and falls below it in Taif by a margin that is not resolved. (C5) A clairvoyant optimality bracket, from the best attained schedule no warmer than the baseline to a provable relaxation bound, with the bound also reported for every arm at its own mean temperature. (C6) A three-layer report of every result (on-commands avoided, electricity, and money as a range over the supported charge per on-command), which shows that at matched mean temperature the cost difference is the on-command charge alone; and a Riyadh check of whether the single-zone ordering survives in dwellings of two and four rooms with independent units.

Section 2 reviews the literature. Sections 3 and 4 define the thermal and cost models and formalise the constrained problem and its optimality conditions; Section 5 sets out the controllers, Section 6 the three climates and Section 7 the protocol. Section 8 reports the results, and Section 9 compares them with published figures and gives the ablation, sensitivities, actuation step and scalability. Section 10 assesses the verification, validity and generalisability of the findings, what governs the saving and the limitations, and Section 11 concludes.

2. Literature Review

The review covers the Saudi context, the control approaches, and binary actuation with multi-zone coupling and tariffs, and closes with the gap this study addresses.

2.1. Saudi Context

Al-Homoud and Krarti [2] attribute the Kingdom's residential demand to a hot climate, low energy awareness and a historically subsidised tariff. Residential electricity is priced on a stepwise schedule: since 2018, 0.18 SAR/kWh up to 6,000 kWh/month and 0.30 above [14], repriced to 0.21 and 0.33 from January 2027 [15]. Compliance with the 2018 Saudi Building Code [16] reduces energy use intensity by up to 20–25% in Climate Zone 2 [17], but leaves plant operation unaddressed, and the rebound effect offsets part of the gains from appliance upgrades [18].

2.2. Control Approaches

The standard residential controller is a fixed-deadband thermostat with no view of tariff, occupancy or weather [19]. MPC optimises a cost over a rolling horizon, typically as a mixed-integer linear programme [20, 21], and reviews document substantial savings over rule-based control, largely under idealised conditions [7]. Its complexity grows with the number of binary variables and zones; for a single zone a dynamic-programming MPC needs only a one-dimensional value grid (Section 9.4) and is used here as the causal model-based comparator.

Most RL studies remain in simulation, with training cost, safety and generalisation the main barriers [22]. Reported savings range from 10 to 22% against rule-based or learned baselines [10, 9, 23, 24]

(Table 5); model-based, multi-agent and transfer variants extend the scope [25, 26, 27], and PPO matches soft actor–critic (SAC) at lower cost in an empirical comparison [28], which is why it is adopted here. The hybrid scheme of [11] imposes hard rules that override the agent near the constraint boundaries.

2.3. Binary Actuation, Multi-Zone Coordination and Tariffs

Most RL-HVAC work targets continuous modulation [29], whereas Saudi residential split units are fixed-capacity and switched on and off, so a charge c_{sw} per on-command stands for start-up wear; schedules that ignore it incur cycling and rebound costs the optimiser does not see [30]. For a single-variable Simple Linear Hybrid Automaton (SLHA) with per-mode running costs and per-switch costs, an optimal infinite-horizon schedule is computable in LogSpace while the finite-horizon problem is NP-complete, with a constant-factor approximation and an FPTAS [13]; the multi-variable extension is in NExpTime, with a PSpace ε-approximation and no hardness result [31]. In multi-zone buildings shared walls couple the zones: GA pre-scheduling with MADDPG reports 6.7% over a rule-based baseline [32], a Markov-game formulation is solved by multi-agent DRL [33], and in non-cooperative distributed MPC each zone minimises its own cost with no plant-wide optimality guarantee [34]. Most scheduling studies assume a constant price [35] or a smooth approximation [20]; Proposition 1 below shows that a block tariff raises the price of the last kWh in a billing month but gives no incentive to rearrange consumption within it.

2.4. Research Gap

Table 1 summarises the positioning of the present work. The nearest neighbour to this study is the present author's own recent work in this journal [12]. That paper addresses binary on/off scheduling for Saudi residential split units under the two-block tariff, with inter-zone thermal coupling, at two building scales, and with a cost-model sensitivity across step-wise, linear and exponential tariffs. It enforces comfort through a soft physics-informed reward and a top-k projection, neither of which excludes a violation, and it establishes no optimality certificate. It names as priority future work a quantification of the remaining optimality gap against a day-ahead mixed-integer scheduler, a longer-horizon model-predictive baseline and modern constrained reinforcement learning, and separately, among its limitations, observes that formal constraint enforcement would provide stronger safety properties than its top-k projection heuristic.

The present study takes up both, though not by the route that paper anticipated. It supplies an optimality certificate, as a provable bound under the model that no schedule at least as cool as the baseline can beat (Section 8.1), and it compares MPC at horizons from 15 min to 48 h under the same protocol as every other arm (Section 8.3). It enforces comfort by construction, under the nominal plant parameters, through a rule layer that overrides the learner's sampled action, a post-posed shield [36], rather than through constrained policy optimisation. It does not quantify that paper's own optimality gap, which concerns a different instance, and it does not re-derive its cost-model or zone-count results, which stand. Its multi-zone evidence is narrower than that paper's: a Riyadh check, with independent per-room units, of whether the single-zone ordering survives in two- and four-room dwellings, without an optimality reference (Section 8.4).

Neither that paper nor the others in Table 1 addresses the following three points, which form the gap this paper addresses: (i) hard comfort enforcement by construction under the nominal plant model, with the price of that guarantee measured; (ii) a clairvoyant optimality bracket against which a tuned rule, a learned policy and MPC are all read; and (iii) evaluation across three Saudi climates under one protocol, in which every arm is selected by the same rule and compared, week by week, with a baseline shifted to its own mean occupied temperature. Table 1 records two qualifications.

3. Thermal and Cost Models

Closing those gaps requires a model in which comfort is a constraint rather than a reward term, and in which the price a schedule pays is the price the meter charges. This section builds that model in three parts: the thermal dynamics of a zone, the topologies over which they are defined, and the binary plant and tariff that price a schedule.

3.1. RC Thermal Model: Heat Loss, Gain, and Inter-Zone Transfer

Each zone $i \in \mathcal{N} = \{1, \ldots, N_z\}$ is modelled as a first-order RC thermal node, the simplest mem-

Table 1: Positioning of the present study relative to selected prior work. ✓= addressed; ○ = partially addressed; – = not addressed.

Study	Binary on/off	Hard comfort	Step-wise tariff	Multi-zone	Saudi climate	Multi-city
[10]	–	○	–	–	–	–
[23]	–	○	–	✓	–	–
[9]	–	○	–	–	–	–
[11]	–	✓	–	–	–	–
[32]	–	○	–	✓	–	–
[12]	✓	○	✓	✓	✓	–
This work	✓	✓	○	✓	✓	✓

The step-wise tariff is formulated (Section 3.4), but the modelled dwelling never crosses a tier boundary under the schedule in force (Remark 3), hence ○. The multi-zone entry for this work covers one city and independent per-room units, without an optimality reference (Section 8.4). The ○ for hard comfort in [12] reflects enforcement by a soft physics penalty rather than by construction.

ber of the RC thermal-network family [37, 38, 39] that is the standard control-oriented approach for residential buildings. Di Natale et al. [40] demonstrated that physically consistent neural RC models scale to multi-zone settings while preserving interpretability; the present work retains the classical first-order RC for transparency and reproducibility. Let $C_{\mathrm{th}}^{(i)}$ (kJ/K) be the effective thermal capacity of zone i, a single lumped value covering the room air together with its furnishings and the participating inner layers of the fabric (Remark 1), and $T_i(t)$ the temperature of that node; comfort is evaluated on it, so it stands for air and participating fabric together. The continuous-time energy balance for zone i is:

$$C_{\mathrm{th}}^{(i)}\frac{\mathrm{d}T_i}{\mathrm{d}t} = \underbrace{-Q_{\mathrm{hvac}}^{(i)}u_i(t)}_{\text{active cooling}}\underbrace{-\Lambda_i(T_i-T_{\mathrm{out}})}_{\text{envelope}} + \underbrace{\sum_{j\in\mathcal{N}_i}\lambda_{ij}(T_j-T_i)}_{\text{inter-zone transfer}} + \underbrace{Q_{\mathrm{sol}}^{(i)}(t)}_{\text{solar}}, \tag{1}$$

where $\Lambda_i = \lambda_{\mathrm{ext}}^{(i)} + \lambda_{\mathrm{win}}^{(i)}$ (W·K^{-1}) is the total envelope conductance, $u_i(t) \in \{0,1\}$ is the binary HVAC state, and $\mathcal{N}_i$ is the set of adjacent zones.

Within one actuation interval the exogenous forcing (outdoor temperature, solar irradiance and the binary command) is held constant, so (1) is a linear system with constant inputs over the interval and its solution is available in closed form. Writing the system in deviation form $\dot{\mathbf{T}} = \mathbf{M}\left(\mathbf{T}-\mathbf{T}_\infty^{(t)}\right)$ with

$$\mathbf{M} = -\tfrac{1}{C_{\mathrm{th}}}\left[\mathrm{diag}(\Lambda_i+\Lambda_{\mathrm{iz},i}) - [\lambda_{ij}]\right], \qquad \mathbf{T}_\infty^{(t)} = -\mathbf{M}^{-1}\mathbf{f}^{(t)}, \tag{2}$$

where $\mathbf{f}^{(t)}$ collects the forcing $C_{\mathrm{th}}^{-1}\left(\Lambda_i T_{\mathrm{out}}^{(t)} + Q_{\mathrm{sol}}^{(i,t)} - Q_{\mathrm{hvac}}^{(i)}u_i^{(t)}\right)$, the exact update over an interval of length Δt is

$$\mathbf{T}^{(t+1)} = \mathbf{T}_\infty^{(t)} + e^{\mathbf{M}\Delta t}\left(\mathbf{T}^{(t)} - \mathbf{T}_\infty^{(t)}\right), \tag{3}$$

which for a single uncoupled zone reduces to the scalar form used throughout the single-zone analysis,

$$T^{(t+1)} = T_\infty^{(t)} + \left(T^{(t)} - T_\infty^{(t)}\right)e^{-\Delta t\,\Lambda_i/C_{\mathrm{th}}}, \qquad T_\infty^{(t)} = T_{\mathrm{out}}^{(t)} + \frac{Q_{\mathrm{sol}}^{(t)} - Q_{\mathrm{hvac}}u^{(t)}}{\Lambda_i}. \tag{4}$$

Equation (3) is used in preference to forward Euler, for three reasons. First, since $-\mathbf{M}$ is a weighted graph Laplacian plus the strictly positive diagonal $\Lambda_i/C_{\mathrm{th}}$, hence symmetric positive definite, every eigenvalue of $e^{\mathbf{M}\Delta t}$ lies in $(0,1)$ for *every* $\Delta t > 0$: the update is unconditionally stable and no step-size stability condition arises. Forward Euler is stable only for $\Delta t < 2C_{\mathrm{th}}/\mu_{\max}$, with $\mu_{\max}$ the largest eigenvalue of $\mathrm{diag}(\Lambda_i) + [\Lambda_{\mathrm{iz},i}\delta_{ij} - \lambda_{ij}]$; at an air-only capacity of 77.2 kJ/K an hourly step meets it for the single zone ($2C_{\mathrm{th}}/\Lambda_i = 1.8$ h) but not for the coupled topologies (Supplementary Section S1), so for the single zone the issue is accuracy and actuation, not stability. Second, the solution relaxes monotonically towards $\mathbf{T}_\infty^{(t)}$ within the interval, so for a single zone the trajectory's extremes are its endpoints and comfort can be checked exactly on the actuation grid; for coupled zones the check is on the grid only. Third, the step size affects only how often the controller may act, not the fidelity of

the physics; the remaining restriction is the actuation condition (8), a property of the controller.

Inter-zone heat flux and effective inter-zone temperature.. The inter-zone coupling term in (3) is driven by a pairwise heat flux:

$$Q_{ij}^{(t)} = \lambda_{ij}\left(T_j^{(t)} - T_i^{(t)}\right), \tag{5}$$

positive when zone j is warmer. With net load $Q_{\mathrm{iz},i}^{(t)} = \sum_j Q_{ij}^{(t)}$, total inter-zone conductance $\Lambda_{\mathrm{iz},i} = \sum_j \lambda_{ij}$ and the *effective inter-zone temperature*

$$T_{\mathrm{iz},i}^{(t)} = \frac{\sum_{j\in\mathcal{N}_i} \lambda_{ij}\, T_j^{(t)}}{\Lambda_{\mathrm{iz},i}}, \tag{6}$$

the net inter-zone load simplifies to $Q_{\mathrm{iz},i}^{(t)} = \Lambda_{\mathrm{iz},i}\left(T_{\mathrm{iz},i}^{(t)} - T_i^{(t)}\right)$, exactly mirroring the envelope term $\Lambda_i(T_i^{(t)} - T_{\mathrm{out}}^{(t)})$. In a multi-zone instance this variable is the natural state input for anticipating inter-zone heat cascades; the multi-zone MPC of Section 8.4 uses its neighbours' planned temperatures in this form, and no learned policy observes it here.

The RC model serves both as the simulation environment and as the prediction model of MPC (Section 5.5); its state is available to the PPO agent through the state vector (Section 5.2), but the trained policy does not use it to plan. Table 2 states the assembly transmittances and areas from which the conductances are formed. The opaque-wall transmittance is at the SBC 2018 prescriptive maximum [16], $U_{\mathrm{wall}} = 0.45\,\mathrm{W\,m^{-2}K^{-1}}$, and the window transmittance $U_{\mathrm{win}} = 2.8\,\mathrm{W\,m^{-2}K^{-1}}$ is assumed for double glazing, giving a fabric conductance of 11.90 W/K per exposed face; Section 6.2 relates them to the code. Infiltration is carried explicitly at 0.6 air changes per hour and evaluated per city from station pressure, contributing 10.77–12.72 W/K (about one face of fabric). Compliance is mandatory and enforced through municipal permitting and, since October 2024, the activation of the electricity connection [41].

The occupied comfort band is $[T_{\min}, T_{\max}]$=[22, 24] °C; its provenance is set out in Section 6.2.

Remark 1 (Thermal capacity, the actuation condition, and stability)**.** *The RC model* (3) *follows the control-oriented identification framework of Bacher and Madsen [42] and captures sensible heat only. The capacity is* $C_{\mathrm{th}} = 4{,}800\,kJ/K$*, an areal heat capacity of* $50\,kJ\,m^{-2}K^{-1}$ *over* $96\,m^2$ *of participating surface, within the ISO 13786 range for a furnished heavy-construction room; with* $\Lambda_i = 23.75\,W/K$ *the time constant is* $C_{\mathrm{th}}/\Lambda_i = 56\,h$*. ISO 13786's areal capacity is a* 24 *h quantity, and the capacity effective at the period of the baseline thermostat's cycle (THERM, Section 5.5; about* 6 *h) is lower, so the effective capacity probably lies below the reference value, in the range where the tuned thermostat gains least and MPC's margin over it is largest; the reference results therefore probably favour the tuned thermostat, and they are repeated at* 2,400, 7,200 *and* 9,600 *kJ/K (Section 8.3).*

The actuation interval is constrained twice over. Holding the zone inside the band across one interval requires the plant's delivery over that interval to be no larger than the band:

$$\frac{Q_{\mathrm{hvac}}\,\Delta t}{C_{\mathrm{th}}} \;\le\; T_{\max} - T_{\min}. \tag{7}$$

This follows from the energy balance and is invariant to the integration scheme*: subdividing the interval delivers the same energy and the same excursion, so a finer integrator cannot remedy a violation. Condition* (7) *is, however, only necessary. A controller switching on a hysteresis deadband must complete each excursion inside* its own *deadband* δ_{db}*, and the deadband together with those excursions must in turn fit inside the comfort band, since the zone keeps moving for one interval after each switch is commanded. With* ϵ_{hot} *the warming excursion of one interval at peak load* (19)*, both requirements give*

$$\frac{Q_{\mathrm{hvac}}\,\Delta t}{C_{\mathrm{th}}} \;\le\; \delta_{\mathrm{db}} \;\le\; (T_{\max}-T_{\min}) - \frac{Q_{\mathrm{hvac}}\,\Delta t}{C_{\mathrm{th}}} - \epsilon_{\mathrm{hot}}, \tag{8}$$

and for a deadband centred on the band the upper limit tightens to $2\min\!\left(\tfrac{1}{2}(T_{\max} - T_{\min}) - Q_{\mathrm{hvac}}\Delta t/C_{\mathrm{th}},\ \tfrac{1}{2}(T_{\max}-T_{\min}) - \epsilon_{\mathrm{hot}}\right)$*. At the parameters of Table 2 the admissible interval is* $0.375 \le \delta_{\mathrm{db}} \le 1.25\,°C$*, against the* $2\,°C$ *that the band width alone would allow. The distinction matters: a* $1.5\,°C$ *deadband satisfies the band-width bound yet leaks* 0.132 *occupied degree-hours, so the looser form cannot adjudicate the admissibility sweeps it is used for. The THERM baseline of Section 5.5 switches at* $T_{\max} - 0.5$ *and* $T_{\min} + 0.5$*, so* $\delta_{\mathrm{db}} = 1\,°C$ *on the* $[22, 24]\,°C$ *band: inside the interval admissible at a* 15*-min step and near its upper end (Section 9.2).*

This determines the interval used throughout. At $\Delta t = 15\,min$ the left-hand side is $0.375\,^\circ C$ against $\delta_{\mathrm{db}} = 1\,^\circ C$, a factor of 2.7. At $\Delta t = 1\,h$ it is $1.50\,^\circ C$ and the condition fails: THERM then accumulates 4.47 occupied degree-hours in the first held-out Riyadh week, so the two arms are no longer held to the same constraint. At 30 min, and equally at $2{,}400\,kJ/K$ and 15 min, the upper limit of (8) is $0.99\,^\circ C$ and fails narrowly. With the $690\,W$ peak load of (19), an hourly interval would require $C_{\mathrm{th}} \geq (Q_{\mathrm{hvac}}+690\,\mathrm{W})\Delta t/(T_{\max}-T_{\min}-\delta_{\mathrm{db}}) \approx 9{,}700\,kJ/K$ by (8), and $14{,}400\,kJ/K$ for a deadband centred on the band as THERM's is, against the $3{,}600\,kJ/K$ that (7) alone would suggest, so it is inadmissible here and $\Delta t = 15\,min$ is adopted. The admissibility sweeps over interval and capacity and the validation literature for the RC form are in Supplementary Section S1 and Table S1; the dependence of the saving on the interval is reported in Section 9.3.

Remark 2 (What scheduling can and cannot change)**.** *One structural property of the plant bounds every claim made below. Summing (3) over zones, each unordered pair $\{i, j\}$ contributes $\lambda_{ij}(T_j - T_i) + \lambda_{ij}(T_i - T_j) = 0$, so the inter-zone terms cancel exactly: conduction between zones redistributes heat but adds none. Averaging over a periodic steady state, in which the mean of $\mathrm{d}T_i/\mathrm{d}t$ vanishes, leaves*

$$Q_{\mathrm{hvac}}\,\langle u_i\rangle = \Lambda_i\big(\langle T_{\mathrm{out}}\rangle - \langle T_i\rangle\big) + \langle Q_{\mathrm{sol}}^{(i)}\rangle, \qquad (9)$$

so the duty cycle, and hence the daily electrical energy $E_{\mathrm{day}} = 24\,P_{\mathrm{hvac}}\langle u_i\rangle$, is determined by the mean zone temperature alone, given the weather. With $\Lambda_i = 23.75\,W/K$ for the single-face Riyadh zone and a fixed COP, $\partial E_{\mathrm{day}}/\partial\langle T_i\rangle = -0.191\,kWh/day/K$.

This constrains the interpretation of the results in three ways. First, a controller confined to the $2\,^\circ C$ occupied band can move $\langle T_i\rangle$ by at most $1\,K$ from the band centre, bounding the energy it can save relative to a thermostat there at about $0.19\,kWh/day$ (0.38 across the full band). Second, with a constant COP and an occupied band that never changes, pre-cooling lowers *the mean temperature and therefore* raises *consumption; an occupancy setback, which changes the band in time, or a temperature-dependent COP would alter this. Third, a cost comparison between two controllers at different mean temperatures is largely a restatement of (9), which is why every arm in Section 8 is compared with a baseline shifted to its own mean occupied temperature (Section 7).*

At equal mean temperature every arm therefore uses the same terminal-priced energy, up to the finite-window and non-periodic residual of the identity, and the saving equals c_{sw} times the on-commands avoided. With a constant COP and no start-up energy this is an identity of the model, not a finding: the cost saving reported here is a charge on avoided on-commands, assumed to stand for deferred wear, neither an energy saving nor a reduction in the electricity bill, and the energy and on-command columns are reported beside it.

3.2. Zone Configurations

Three topologies are defined. The headline study evaluates the single zone; Section 8.4 evaluates two- and four-room instances in Riyadh; the topologies are defined in the repository code (Section 16). The *single zone* (1×1) has one windowed exterior wall and no inter-zone coupling. The *linear array* ($1\times N$, $N \in \{2, 4, 6\}$) places N rooms in a row, each with one windowed exterior wall (azimuths in Table S-MZ); interior rooms couple to two neighbours. The *grid* ($N \times N$, $N \in \{2, 3, 4\}$) produces up to $N_z = 16$ zones; corner zones have two windowed exterior walls, edge zones one.

3.3. Binary HVAC Model and Interval Cost

Each zone has an independent split-unit air conditioner with two modes: *on* ($u_i = 1$) or *off/idle* ($u_i = 0$, at zero operational cost). The interface is supervisory: $u_i = 1$ commands the unit to deliver its full sensible capacity for the 15-min step, as a smart plug or an infrared command at a low setpoint would, and the model lumps air and mass in one node, so the unit's own return-air cycling is not represented. A charge c_{sw} (SAR) is incurred on each $0 \to 1$ transition, an *on-command*, standing for start-up wear. Whether each on-command corresponds to a compressor start depends on the equipment's own thermostat and is not established here, so the events counted and priced throughout are on-commands. Switching off carries no charge.

Definition 1 (Interval cost)**.** *The cost of zone i at interval t under model m is:*

$$c_i^{(t)} = c_{\mathrm{sw}} \cdot \mathbf{1}\big[u_i^{(t)} > u_i^{(t-1)}\big] + f^m\big(u_i^{(t)}, E_{\mathrm{cum}}^{(t)}\big) \cdot \Delta t, \qquad (10)$$

where $E_{\mathrm{cum}}^{(t)} = \sum_{\tau \leq t}\sum_i P_{\mathrm{hvac}} u_i^{(\tau)} \Delta t$ is cumulative billing-month energy (kWh) up to interval t.

Table 2: Zone parameters for each $4 \times 4 \times 4$ m room.

Symbol	Description	Value	Unit
C_{th}	Thermal capacity (furnished; ISO 13786, Remark 1)	4,800	kJ/K
U_{wall}	Opaque-wall transmittance (SBC 2018 prescriptive max)	0.45	$\text{W}\cdot\text{m}^{-2}\text{K}^{-1}$
A_{wall}	Opaque area per exterior wall	14.0	m^2
U_{win}	Window transmittance (assumed, double glazing)	2.8	$\text{W}\cdot\text{m}^{-2}\text{K}^{-1}$
A_{gl}	Glazing area per exterior wall	2.0	m^2
$\lambda_{\text{ext}} + \lambda_{\text{win}}$	Fabric conductance *per exposed face*	11.90	$\text{W}\cdot\text{K}^{-1}$
Λ_{inf}	Infiltration conductance at 0.6 ACH (per city)	10.77–12.72	$\text{W}\cdot\text{K}^{-1}$
SHGC	Solar heat-gain coefficient	0.25	—
E_{base}	Non-HVAC monthly household load	800	kWh/month
λ_{ij}	Inter-zone shared-wall conductance	12.5	$\text{W}\cdot\text{K}^{-1}$
Q_{hvac}	HVAC sensible extraction (assumed)	2.0	kW
COP	Coefficient of performance (assumed, constant)	2.985	—
P_{hvac}	HVAC electrical draw	0.670	kW
$T_{\min}/T_{\max}$	Occupied comfort band	22/24	°C
δ_{db}	THERM deadband (baseline, Section 5.5)	1.0	°C
$\epsilon_{\text{hot}}/\epsilon_{\text{cold}}$	Rule guard bands, from (19)	0.20/0.40	°C
c_{sw}	Charge per on-command (computational reference)	0.15	SAR
Δt	Actuation interval	15	min

Envelope conductance is affine in the number of exposed faces n, $\Lambda_i(n) = 11.90\,n + \Lambda_{\text{inf}}$, because infiltration is a property of the zone volume. Λ_{inf} depends on station pressure: 11.85 W/K at Riyadh (94.3 kPa), 12.72 at Jeddah (101.2 kPa) and 10.77 at Taif (85.7 kPa, 1,478 m), so the single-face totals are $\Lambda_i = 23.75$, 24.62 and 22.67 W/K.

3.4. Electricity Cost Models

Model L (linear) charges a constant rate, the standard assumption in scheduling [35]: $f^{\text{L}} = c_r^{\text{L}} P_{\text{hvac}} u_i$. **Model E** (exponential), $f^{\text{E}} = c_r^{\text{E}} e^{\beta E_{\text{cum}}} P_{\text{hvac}} u_i$ with $\beta = 5 \times 10^{-4}\,\text{kWh}^{-1}$, is a smooth approximation to step-wise pricing over a comparable price range. Neither is used in any result.

Model S (Step-wise Cost Model). Model S is the step-wise (piecewise-constant) cost model. It is instantiated with the Saudi residential tariff in force since January 2018 [14], a two-block schedule in which the marginal price rises once, at 6,000 kWh per billing month. The billed quantity E_{cum} is the whole-house monthly cumulative consumption (the non-HVAC background load of Section 7 plus HVAC); VAT and the fixed charge are excluded as they do not affect scheduling:

$$p(E_{\text{cum}})\ [\text{SAR/kWh}] = \begin{cases} 0.18, & E_{\text{cum}} \le 6{,}000, \\ 0.30, & E_{\text{cum}} > 6{,}000. \end{cases} \tag{11}$$

From January 2027 the two blocks are repriced to 0.21 and 0.33 SAR/kWh with the same boundary [15] (Remark 3).

Proposition 1 (The bill does not depend on timing)**.** *Let the marginal price be any locally integrable function $p(\cdot)$ of cumulative billing-month consumption alone, and let E_{cum} be non-decreasing and absolutely continuous in t. Then the energy bill over the billing month is $B = \int_0^{E_{\text{total}}} p(e)\,\mathrm{d}e$, which depends only on E_{total} and not on the order in which it was consumed.*

Proof. Charging the rate in force as consumption accumulates gives $B = \int_0^T p\big(E_{\text{cum}}(t)\big)\dot{E}_{\text{cum}}(t)\,\mathrm{d}t$. Absolute continuity makes $e = E_{\text{cum}}(t)$ a valid change of variable and non-decrease makes it orientation-preserving, so $B = \int_0^{E_{\text{total}}} p(e)\,\mathrm{d}e$. No property of p beyond local integrability is used; in particular monotonicity is not required. □

On the actuation grid the realised bill is path-dependent only when a tier boundary falls inside one interval's increment, at the order of one interval's consumption (Supplementary Section S1; no boundary is approached here, Remark 3).

Three consequences follow. First, Models L, E and S are all of this form (constant, exponential in E_{cum}, and a step function of E_{cum}), so none of them can be distinguished by any experiment that rearranges consumption in time. What separates them is the exchange rate $c_{\text{sw}}/(pP_{\text{hvac}}\Delta t)$ between an avoided switch and an hour of running: a constant under Model L, and under Models E

and S a quantity that varies with E_{cum} through the billing month, by a factor of only 1.04 under Model E between the background load and this account's monthly consumption. Together they induce a one-parameter family of behaviours whose range Section 9.2 sweeps, and a separate cost-model ablation would resolve nothing those sweeps do not. Second, pre-cooling is dominated under *every* such tariff, not only under a constant price: it changes when energy is drawn and, by identity (9), raises how much, so the timing earns nothing and the extra energy costs. Third, the proposition and its second consequence fail under different conditions. The *proposition* fails for any price that is not a function of cumulative consumption alone (a demand or capacity charge on coincident power, time-of-use and real-time pricing) and, trivially, across a billing boundary, since the tiers reset monthly and shifting load between months is a genuine arbitrage. The second *consequence*, that pre-cooling is dominated, fails under two further conditions that leave the proposition intact: a minimum on- or off-time constraint, which makes the cost of a given duty depend on its arrangement, and a temperature-dependent coefficient of performance, which is a property of the plant, not of the tariff, and which would make cooling genuinely cheaper at night. None of these is evaluated here (Section 10.4).

Two properties of Model S matter. First, the marginal price is a non-decreasing step function of cumulative consumption, so the bill is convex and representable in a linear programme; the difficulty lies in the binary actions and the switching term, not the tariff. Second, under Model S the per-step cost rate depends on the accumulated state E_{cum}, which takes the problem outside the SLHA class of Section 4.2 and would require E_{cum} in the agent state; under the tariff in force it does not (Section 5.2).

Remark 3 (The tariff currently in force, and what it implies here)**.** *The residential tariff in force from January 2018 to December 2026 is a two-block schedule:* 0.18 *SAR/kWh up to* 6,000 *kWh/month and* 0.30 *SAR/kWh above [14]. From 1 January 2027 the blocks are repriced to* 0.21 *and* 0.33 *SAR/kWh and a fixed monthly charge of* 70–100 *SAR, graded by breaker capacity, replaces the former meter reading, maintenance and billing fee [15]. Both have the single boundary of* (11)*. The fixed charge is schedule-independent and does not alter the optimal policy, but it enlarges the bill; it does not change any percentage here, which is relative to the room's cooling energy cost plus on-command charges, and against the household bill every saving would be far smaller.*

The modelled account consumes ≈884 *kWh per billing month (Section 7), far below the boundary, so it stays in the first block throughout and faces a* constant *marginal price. The scheduling problem therefore belongs to the same cost class as Model L, though not the same instance: Model L is specified at* 0.12 *SAR/kWh (no result uses it) against a first-block rate of* 0.18 *(*0.21 *from 2027). Because* c_{sw} *is a fixed charge that does not scale with the energy price, the ratio* $c_{\text{sw}}/(p\,P_{\text{hvac}}\,\Delta t)$*, which determines the optimal switching policy, falls by a third between those rates. Every arm is therefore reported at the rate in force, and the tuned thermostat (TUNED, Section 5.5) and the bound are repeated at the announced 2027 rate (Table 7(a); MPC is not). A dwelling that crossed the* 6,000 *kWh boundary would face a higher marginal price, which by Proposition 1 leaves the optimal schedule unchanged and lowers the percentage saving.*

The saving therefore arises from longer, less frequent on-phases, that is, from the switching condition (O3 of Section 4.4), not from the energy condition (O2).

Running example. A single-zone instance (1×1, Riyadh, July, $T_{\text{out}} = 40\,^{\circ}\text{C}$) at the parameters of Table 2 is used throughout to fix intuition about the cycle structure of Observation 2; Fig. 1 draws it from the closed-form scalar solution. ◇

With the thermal and cost models established, the following section formalises the constrained optimisation and its optimality conditions.

4. Optimal Schedule: Constrained Formulation and Optimality Conditions

This section casts the problem over an infinite horizon (Section 4.1), relates its single-zone core to simple linear hybrid automata (Section 4.2), states the monthly constrained problem (Section 4.3) and derives the optimality conditions and a corollary on the rule layer (Section 4.4).

4.1. Infinite Time Horizon and Limit-Average Cost

A residential HVAC system operates indefinitely: months repeat and no planning horizon ends the building's operation, so the relevant cost measure is the long-run average cost per interval. Following

Mousa et al. [13], the problem is cast as an *infinite time horizon* optimisation whose performance criterion is the *limit-average cost*.

Definition 2 (Infinite schedule and limit-average cost)**.** *An* infinite schedule *is the binary matrix* $\sigma = (u_i^{(t)})_{N_z \times \infty}$, $u_i^{(t)} \in \{0,1\}$. *Its limit-average cost under model* m *is:*

$$\bar{J}_\infty^m(\sigma) = \limsup_{H\to\infty} \frac{1}{H} \sum_{t=0}^{H-1} \sum_{i\in\mathcal{N}} c_i^{(t)}, \qquad (12)$$

where $c_i^{(t)}$ *is the interval cost of Definition 1. A schedule is* safe *if, for all* i *and all* $t \geq 0$, $T_i^{(t+1)} \leq T_{\max}$ *on every occupied step and* $u_i^{(t)} = 1$ *implies* $T_i^{(t+1)} \geq T_{\min}$*: the ceiling is pointwise and a cooling step may never end below* $T_{\min}$*, while a free-floating step* ($u_i^{(t)} = 0$) *has no lower limit.*

The criterion is defined over an infinite horizon, but every number reported is computed on a finite one: each schedule is scored over 168 h windows of $N = 672$ steps, MPC re-plans every step over a lookahead of 0.25–48 h, and the clairvoyant schedule and the bound treat each window as a whole, with the terminal value V_N (Section 5.5) standing in for the continuation so that the end of a window is not exploited.

4.2. Simple Linear Hybrid Automata: Formal Connection

The SLHA connection [13] provides a formal language for *complete cycles*, establishes that an optimal infinite-horizon schedule for the single-zone constant-tariff problem is computable in LogSpace, and identifies which features of the full problem take it outside that tractable class.

Definition 3 (Single-zone SLHA)**.** *The single-zone binary HVAC system under Model L is the SLHA* $\mathcal{A} = (Q, X, \mathrm{Inv}, f, \Delta, \mathcal{C})$ *with modes* $Q = \{q_0$ *(idle)*$, q_1$ *(cool)*$\}$*, variable* $X = \{T\}$*, flows* $\dot{T}{=}a_0 \triangleq (\Lambda(T_{\text{out}} - T) + Q_{\text{sol}})/C_{\text{th}}$ *in* q_0 *and* $\dot{T}{=}a_1 \triangleq a_0 - Q_{\text{hvac}}/C_{\text{th}}$ *in* q_1 *(*$a_1 < 0 < a_0$ *in Saudi summer), invariants* $T \leq T_{\max}$ *in* q_0 *and* $T \geq T_{\min}$ *in* q_1*, transitions* $(q_0 \to q_1)$ *on* $T \geq T_{\max} - \epsilon_{\text{hot}}$ *with cost* c_{sw} *and* $(q_1 \to q_0)$ *on* $T \leq T_{\min} + \epsilon_{\text{cold}}$ *with cost* 0*, and continuous cost rate* $c_r P_{\text{hvac}}$ *in* q_1 *only.*

A state diagram of the automaton is in the repository (Section 16).

The rates of Definition 3 depend on T through $\Lambda(T_{\text{out}} - T)$, so reading them as constants is an abstraction; at $\tau = 56$ h its error in the band-crossing time is 0.1–0.2% (Supplementary Section S2), and every numerical result is computed on the exact dynamics (3).

Multi-zone coupling, time-varying disturbances and Model S's consumption-dependent cost rate each take the problem outside the SLHA class, since the LogSpace result of [13] requires a single continuous variable, a time-invariant automaton and a constant per-mode cost rate. No polynomial-time algorithm is known for the extended problem, and the finite-horizon single-variable case is already NP-complete [13]; no hardness result is claimed for the multi-zone problem, which remains open.

Definition 4 (Leap and complete cycle)**.** *A* leap *is a maximal contiguous time interval during which the automaton stays in one mode. An* on-leap ℓ^+ *(in mode* q_1*) of duration* τ^+ *cools the zone from* $T_{\max}$ *to* $T_{\min}$ *at rate* $|a_1|$*:* $\tau^+ = (T_{\max} - T_{\min})/|a_1|$*. An* off-leap ℓ^- *(in mode* q_0*) of duration* τ^- *returns temperature from* $T_{\min}$ *to* $T_{\max}$ *at rate* a_0*:* $\tau^- = (T_{\max} - T_{\min})/a_0$*. The on-command charge* c_{sw} *is incurred at the start of each on-leap only. A* complete cycle *is one on-leap followed by one off-leap. Its total cost and duration are:*

$$\mathcal{C}(\ell^+, \ell^-) = c_{\text{sw}} + c_r P_{\text{hvac}}\, \tau^+, \quad \tau_{\text{cyc}} = \tau^+ + \tau^-. \qquad (13)$$

The resulting limit-average cost per unit time *of repeating complete cycles indefinitely is:*

$$\bar{c}^* = \frac{c_{\text{sw}} + c_r P_{\text{hvac}}\, \tau^+}{\tau^+ + \tau^-}. \qquad (14)$$

Observation 1 (Optimal infinite schedule for SLHA, Mousa et al. [13])**.** *The optimal safe infinite schedule for the single-zone SLHA of Definition 3 is to repeat the cheapest complete cycle* (ℓ^+, ℓ^-) *indefinitely. Its limit-average cost equals* $\bar{c}^*$ *of equation (14), which is achievable in deterministic LogSpace (Theorem 1 of [13]).*

The SLHA analysis is not used as a solution method for the full problem but as a diagnostic tool that identifies the structural properties on which the controller design draws. It yields three design inputs: (i) optimal schedules have complete-cycle structure (Observation 2), motivating rules R1–R2; (ii) the ratio τ^+/τ^- governs cycle structure and motivates the one-hour outdoor forecast

in the agent state; (iii) the absence of any known polynomial-time algorithm for the multi-zone problem [31] motivates approximate controllers, learned or receding-horizon. No controller here solves the monthly problem (15) (Section 4.3) itself. That problem states the objective, matched to the tariff's monthly reset; the controllers approximate it on a receding basis, and it is evaluated on 168 h windows.

The rule layer is weaker than the structural result that motivates insight (i). Observation 2 characterises *optimal* single-zone schedules as sequences of complete cycles. Rules R1 and R2 of Definition 7, by contrast, are boundary-triggered overrides: they fix the action only on $T_i \geq T_{\max} - \epsilon_{\text{hot}}$ and on $T_i \leq T_{\min} + \epsilon_{\text{cold}}$, and leave it unconstrained on the interior $(T_{\min} + \epsilon_{\text{cold}},\, T_{\max} - \epsilon_{\text{hot}})$, which for the guard bands of (19) is 1.40 K of the 2 K band. They impose no minimum dwell and never require a cycle, once begun, to complete, so a schedule switching every step inside the interior satisfies them exactly with no complete-cycle structure at all. The rule layer thus enforces O1 by construction for the nominal plant parameters from which (19) derives its margins (Section 5.3 states the limits of that guarantee) and is *consistent with* the structure of Observation 2 without *implementing* it; complete cycles are left to the learned policy's switching term, with no guarantee. A rule set that did implement it (a minimum-dwell constraint) is identified as required work in Section 10.4.

4.3. Constrained Optimisation Problem

Definition 5 (Schedule over one billing month)**.** *A* monthly schedule *over horizon* $\mathcal{H} = \{0, \ldots, H-1\}$ *(with* $H = 2{,}880$ *intervals of* 15 *min for a 30-day month) is the binary matrix* $\sigma = (u_i^{(t)})_{N_z \times H}$, $u_i^{(t)} \in \{0, 1\}$.

Definition 6 (Constrained optimisation problem)**.** *The monthly scheduling problem is:*

$$
\begin{aligned}
\min_{\sigma \in \{0,1\}^{N_z \times H}} \bar{J}^m(\sigma) &= \frac{1}{H} \sum_{t \in \mathcal{H}} \sum_{i \in \mathcal{N}} c_i^{(t)} \\
\text{s.t.} \quad & T_i^{(t+1)} \leq T_{\max} \quad \forall\, i,\, t \in \mathcal{H}_{\text{occ}} \\
& \textit{(ceiling, } \textbf{\textit{hard}}\textit{)} \\
& u_i^{(t)} = 1 \Rightarrow T_i^{(t+1)} \geq T_{\min} \quad \forall\, i,\, t \in \mathcal{H} \\
& \textit{(cold rule, } \textbf{\textit{hard}}\textit{)} \\
& T_i^{(t+1)} = f\Big(T_i^{(t)}, u_i^{(t)}, \{T_j^{(t)}\}_{j \in \mathcal{N}_i}, T_{\text{out}}^{(t)}\Big) \\
& \textit{(RC dynamics, eq. (3))}
\end{aligned}
\tag{15}
$$

Here $\mathcal{H}_{\text{occ}} \subseteq \mathcal{H}$ *is the set of occupied steps. A schedule satisfying both comfort constraints is* safe *(Definition 2). Both are* hard*: no cost saving justifies an occupied step ending above* $T_{\max}$ *or a cooling step ending below* $T_{\min}$*. A free-floating step has no lower limit, because a cooling-only plant cannot stop a zone drifting below* $T_{\min}$ *when the outdoor air is cool; the protocol (Section 7) reports such free-float under-temperature and gives it no credit in the clipped mean, rather than counting it as a violation of the schedule. On away steps, used only by the setback arms, the ceiling is the empty-house limit of Section 8.2. The RC dynamics embed four heat-transfer mechanisms (active cooling, envelope loss/gain, solar heat gain, and bidirectional inter-zone transfer), making this a multi-dimensional, non-linear, constrained binary programme. The inter-zone term* $\sum_{j \in \mathcal{N}_i} \lambda_{ij}(T_j - T_i)$ *is embedded directly in the RC constraint (3): heat conducted from a warmer neighbour raises* $T_i^{(t+1)}$*, potentially triggering the ceiling constraint in zone* i *even when* $u_i^{(t)} = 0$*. An optimal schedule must therefore coordinate all zones simultaneously.*

4.4. Optimality Conditions

A safe schedule σ^* is *optimal* if it satisfies O1 and resolves the O2–O3 trade-off at least cost, where the three conditions are: **(O1) Comfort feasibility**: $T_i^{(t)} \leq T_{\max}$ for all i and every occupied t, and no cooling step ends below $T_{\min}$; both are hard, and free-float drift below the band is measured (Definition 6). **(O2) Minimum energy at the prevailing marginal price**: the schedule draws as little energy as the comfort constraint permits. Under every cost model used here this is the whole of the economic content of running time, because Proposition 1 shows the bill depends on total consumption and not on its arrangement; there is no "cheap-

est window" to concentrate into. **(O3) Minimum switching frequency**: on-phases are as long and infrequent as the thermal state permits, amortising the discrete cost c_{sw} per activation.

O1 enforces the hard constraint; O2 and O3 are the two economic terms of $\bar{J}^m$ in equation (15), and they can trade off. At a fixed mean temperature the energy term is fixed by identity (9), so what separates schedules compared at equal mean is O3 (Section 8.2). In the infinite-horizon interpretation (12), an optimal schedule repeats a cost-minimising monthly pattern indefinitely. For the single-zone SLHA subproblem this pattern is the cheapest complete cycle (Observation 1). No claim is made that the learned schedules attain it (Section 4.2).

For the running example ($T_{\mathrm{out}} = 40\,^{\circ}\mathrm{C}$, flat price at the in-force 0.18 SAR/kWh), the complete cycle of Fig. 1 gives $\bar{c}^{*} \approx 0.042\,\mathrm{SAR/h}$; switching every 3 h instead, at the same duty ratio and so violating O3, costs 0.074 SAR/h, which is 75% more. That difference is O3's monetary value.

Remark 4 (Switching cost and schedule granularity)**.** *Condition O3 implies that a higher c_{sw} favours fewer, longer on-phases (Observation 2), and therefore that the saving available to any scheduler grows with it. Section 9.2 measures that dependence and Section 10.3 draws the consequence: the reported figure is a property of the tariff as much as of the controller. The guard thresholds do not need to move with c_{sw}, because (19) derives them from the actuation interval and the load, neither of which depends on a price.*

Observation 2 (Complete-cycle structure)**.** *Within the SLHA abstraction of Section 4.2 (single zone, $c_{\mathrm{sw}} > 0$, constant marginal price), an optimal schedule consists of* complete cycles*: an on-phase during which T_1 cools from $T_{\max}$ to $T_{\min}$, followed by an idle phase during which T_1 drifts back to $T_{\max}$. This is Theorem 2 of [13] applied to the abstraction: a schedule that switches on before $T_{\max}$ or off before $T_{\min}$ can be merged into a longer on-phase at the same continuous cost and one fewer switch. The corollary below does not rely on it.*

Fig. 1 draws this structure at the parameters of this study. Observation 2 has a consequence for a learner placed inside the rule layer, under hypotheses that the evaluated problem does not meet; the corollary states it and Remark 5 measures what the gap between the two is worth.

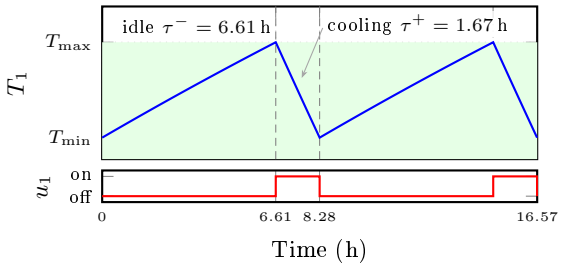


Figure 1: The complete-cycle structure of Observation 2, drawn from the closed-form scalar solution at $T_{\mathrm{out}} = 40\,^{\circ}\mathrm{C}$, $\Lambda_i = 23.75\,\mathrm{W/K}$ and $C_{\mathrm{th}} = 4800\,\mathrm{kJ/K}$. The optimal schedule touches each band edge exactly once per cycle: the plant runs only while cooling $T_{\max} \to T_{\min}$ and is idle while the zone drifts back. The dwell times follow in closed form from the scalar update, $\tau^- = \tau \ln[(T_{\mathrm{out}} - T_{\min})/(T_{\mathrm{out}} - T_{\max})] = 6.61\,\mathrm{h}$ and $\tau^+ = \tau \ln[(T_{\max} - T_{\infty}^{+})/(T_{\min} - T_{\infty}^{+})] = 1.67\,\mathrm{h}$ with $\tau = C_{\mathrm{th}}/\Lambda_i = 56.1\,\mathrm{h}$, giving a cycle of 8.28 h, or 2.90 on-commands per day at this constant 40 °C load; on the held-out Riyadh weeks the clairvoyant schedule issues 2.83 on-commands per day (THERM's 4.10 less the 31.1% of Table 3). Corollary 1 does not depend on this structure; it uses only the bounded widths under constant drift.

Corollary 1 (Optimality of the rule-induced hysteresis under constant disturbance)**.** *Consider the single-zone problem with $c_{\mathrm{sw}} > 0$ under a constant marginal price, with the hard rules of Definition 7 in force,* with T_{out} and Q_{sol} constant, *so that the drift rates a_0, a_1 of Definition 4 are time-invariant, and compare policies* at the mean occupied temperature of the maximal-width cycle, *the midpoint of $\mathcal{F}$ below. Let $\mathcal{F} = (T_{\min} + \epsilon_{\mathrm{cold}}, T_{\max} - \epsilon_{\mathrm{hot}})$ be the interval on which the agent's action is not overridden: the rules fix the action everywhere else, so this is the whole class over which a learned policy can act. Then the rule-induced hysteresis (continue the previous action throughout $\mathcal{F}$) attains cost no greater than that of any policy that differs from it on $\mathcal{F}$.*

Proof. Write $T_{\mathrm{on}}^{g} = T_{\max} - \epsilon_{\mathrm{hot}}$ and $T_{\mathrm{off}}^{g} = T_{\min} + \epsilon_{\mathrm{cold}}$ for the guard thresholds, and $W = T_{\mathrm{on}}^{g} - T_{\mathrm{off}}^{g} = |\mathcal{F}|$ for the width they enclose.

Step 1: phases and widths. Any policy is a sequence of on-phases and off-phases. On-phase k starts at U_k and ends at L_k; off-phase k starts at L_k and ends at U_{k+1}. The rules force the action outside $\mathcal{F}$, so every switch point lies in $[T_{\mathrm{off}}^{g}, T_{\mathrm{on}}^{g}]$ and the widths $w_k = U_k - L_k$ and $v_k = U_{k+1} - L_k$ obey $w_k, v_k \le W$. No merging argument is needed.

Step 2: the energy term agrees across all competi-

tors. With a_0, a_1 constant, on-phase k lasts $w_k/|a_1|$ and off-phase k lasts v_k/a_0. Because the temperature stays in a bounded interval, $|\sum_{k\le n} w_k - \sum_{k\le n} v_k| \le W$ for every n, so every policy has the long-run duty ratio $a_0/(a_0 + |a_1|)$ (identity (9) in its time-invariant form) and the same energy cost per unit time at a constant marginal price. The comparison reduces to the switching term.

Step 3: switch count is minimised by maximal width. Over a horizon H containing $n(H)$ starts, $H = \sum_{k\le n(H)} \big(w_k/|a_1| + v_k/a_0\big) + O(W/a_0)$, so

$$\lim_{H\to\infty} \frac{n(H)}{H} \;=\; \Big[(1/a_0 + 1/|a_1|)\,\overline{w}\Big]^{-1},$$

with

$$\overline{w} = \lim_{n\to\infty} \tfrac{1}{n} \sum_{k\le n} w_k,$$

whenever the limit exists (the same bound holds for the limit superior otherwise). Since $w_k \le W$, $\overline{w} \le W$, with equality if and only if $w_k \to W$ in density, that is, if and only if the policy switches at the guard thresholds on all but a vanishing fraction of cycles, which is the rule-induced hysteresis on the states it visits. Every other policy therefore has a switching frequency, and so a cost, no smaller. □

Under constant drift the maximal-width cycle attains only the midpoint of $\mathcal{F}$ as its mean; at any other prescribed mean narrower cycles must be used and the argument no longer selects the hysteresis. The effect of exponential drift and of the 15-min step on the sampled cycle mean is measured in Supplementary Section S2; matching in the protocol is done on the simulated trajectory, never on the midpoint formula.

Remark 5 (Room inside the rules)**.** *Neither hypothesis of Corollary 1 holds in the protocol. The disturbance is diurnal, and the matched mean lies below the midpoint of the training rule set's free interval $\mathcal{F}$ (23.10 °C for R1/R2 at 23.80/22.40 °C). The guard hysteresis reaches the target only by lowering its own ceiling (fitted R1 23.60–23.66 °C), which narrows $\mathcal{F}$ and leaves the corollary silent about policies in the original $\mathcal{F}$. Section 8.3 measures that room with causal controllers, with the comfort-offset caveat of Section 10.4.*

Within the constant-rate abstraction the duty ratio, and so the energy, is the same for every policy (Step 2). Under the exact, temperature-dependent dynamics it is not: away from matched comfort, switching off early raises the mean temperature, which *reduces* energy by (9) while adding a switch, so the trade can go either way. Matching the mean removes the energy term and leaves the switching term (Remark 2). Because the corollary does not hold under the diurnal drift of the evaluated problem, PPO's coincidence with the guard hysteresis (Section 9.2) does not confirm it.

The corollary is narrow, and its hypotheses locate where a better policy can act: a diurnal disturbance or a time-varying price makes the cost-minimising switch points time-dependent, an occupancy schedule changes the band, and a shared capacity-limited plant couples the zones' action sets.

5. Controllers

Section 4.4 establishes that O1 must hold at every interval while O2 and O3 are economic. The rule-constrained PPO controller (RBRL) splits the controller along exactly that line: a rule layer that enforces O1 by construction under the nominal plant parameters, and a learned layer that works on O2 and O3 inside the region the rules leave free. Its comparators, rule-based, model-based and clairvoyant, are defined in Section 5.5.

5.1. Architecture

Fig. 2 shows the two layers. The rule layer overrides the agent's sampled action whenever the zone temperature reaches a guard threshold. It is applied in the environment after sampling, a post-posed shield [36], so the learner stores its own sampled action and treats the override as part of the dynamics; invalid-action masking inside the policy [43] would remove the forced steps from the gradient but, with two actions, would leave the optimum unchanged. The PPO layer chooses the action where the rules leave it free; Section 9.2 tests what it achieves there.

5.2. State Space

The numerical study controls a single zone, and at interval t the agent observes eight features:

$$s_t = \Big(T^{(t)},\; u^{(t-1)},\; T^{(t)}_{\text{out}},\; T^{(t+4)}_{\text{out}},\; I^{(t)},\; \sin\tfrac{2\pi h_t}{24},\; \cos\tfrac{2\pi h_t}{24},\; t/N\Big) \in \mathbb{R}^8, \tag{16}$$

namely the zone temperature, the previous action, the outdoor temperature now and one hour ahead, the south-vertical irradiance, the hour of day as

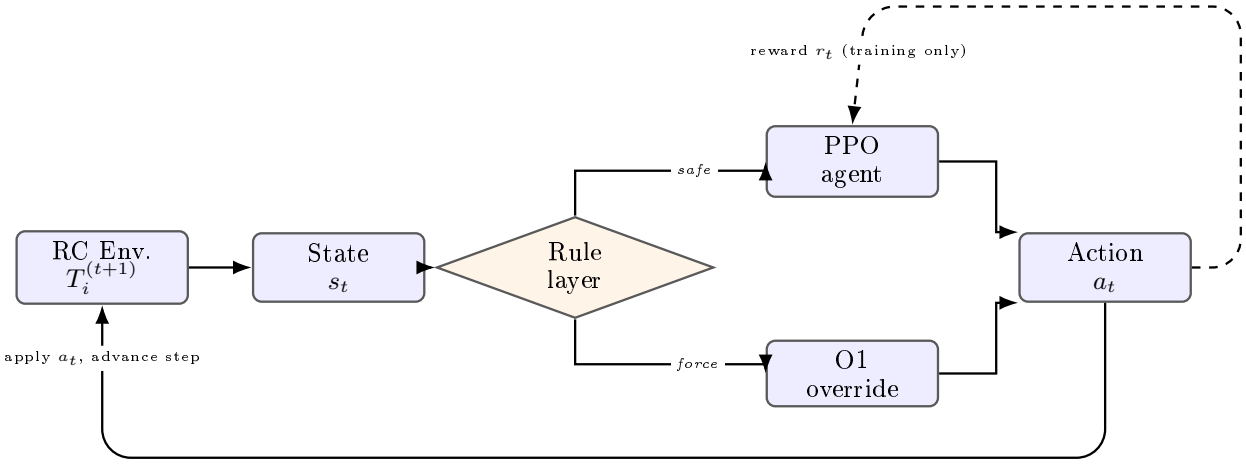


Figure 2: RBRL architecture. The rule layer routes to the PPO agent when no comfort boundary is threatened (*safe*) and overrides it with the O1-enforcing action when one is imminent (*force*); the dashed reward loop is used in training only.

a sine–cosine pair, and the fraction t/N of the episode elapsed, each scaled to order one (scalings in the repository). The one-hour outdoor value is a perfect forecast. The effective inter-zone temperature $T_{\mathrm{iz},i}$ (equation (6)) is not observed, because there is no neighbouring zone, and neither is E_{cum}: under the in-force two-block tariff this dwelling never leaves the first block (Remark 3), so the cost rate does not depend on it. A learned multi-zone or stepped-tariff policy would need both; none is trained here.

5.3. Rules and Action Space

Definition 7 (Rule-based overrides)**.**

$$\textbf{\textit{R1}}\ (\textit{force on}):\ T_i^{(t)} \geq T_{\max} - \epsilon_{\mathrm{hot}} \ \Rightarrow\ u_i^{(t)} \leftarrow 1. \tag{17}$$

$$\textbf{\textit{R2}}\ (\textit{force off}):\ T_i^{(t)} \leq T_{\min} + \epsilon_{\mathrm{cold}} \ \Rightarrow\ u_i^{(t)} \leftarrow 0. \tag{18}$$

The guard bands are *asymmetric* and are derived, not assumed. Each must absorb the temperature excursion of one actuation interval in its own direction:

$$\begin{aligned} \epsilon_{\mathrm{cold}} &\geq \frac{Q_{\mathrm{hvac}}\Delta t}{C_{\mathrm{th}}}, \\ \epsilon_{\mathrm{hot}} &\geq \frac{\Delta t}{C_{\mathrm{th}}} \max_t \left[\Lambda_i (T_{\mathrm{out}}^{(t)} - T_{\max}) + Q_{\mathrm{sol}}^{(i,t)}\right]. \end{aligned} \tag{19}$$

The two differ because cooling is driven by the plant and warming by the load. At the parameters of Table 2 the cooling excursion is 0.375 °C; the peak seasonal load of 690, 664 and 513 W across the three cities gives warming excursions of 0.129, 0.125 and 0.096 °C. Cooling is about 2.9 times the fastest warming, so $\epsilon_{\mathrm{cold}} = 0.40$ and $\epsilon_{\mathrm{hot}} = 0.20$ °C are adopted; the agent then acts on $(22.40, 23.80)$ rather than the $(22.5, 23.5)$ a symmetric 0.5 °C guard would leave. Even so, the rule layer can exclude the tuned thermostat's own switch points (23.71/22.36, 23.70/22.34 and 23.64/22.32 °C in Riyadh, Jeddah and Taif, all with an off point below R2): that exclusion is a property of the method, since any fixed guard band forbids some members of the family it guards. The guarantee holds for the nominal plant only. The cold-side margin of (19) omits the envelope term, which adds to the cooling when the outdoor air is cooler than the zone, so in the coolest in-season hours it is not strict; and it is derived at the reference capacity, so at a lower effective capacity (Remark 1) a cooling step can overshoot it (by 0.75 K per step at 2,400 kJ/K). The realised commanded under-temperature of the rule-layer arms is nevertheless zero in the headline configuration (Section 8).

The rule layer contains no pre-cooling rule, because Proposition 1 and Remark 2 rule it out under every cost model used here. Pre-cooling would pay only under a physical condition, not a tariff one: shifting cooling from t_2 back to t_1 pays only if the price of *delivered* cooling falls by more than the store leaks over the interval, which at $\tau = C_{\mathrm{th}}/\Lambda_i = 56.1$ h is 3.5% over a four-hour shift and 6.8% over eight, averaged over the shift; the

end-point figures $1 - e^{-\Delta t/\tau}$ are 6.9% and 13.3%. A cumulative-block tariff supplies none of that. A temperature-dependent coefficient of performance would supply 15–20% across a typical diurnal swing and is the route by which a pre-cooling rule could be reinstated (Section 10.4). The action is $a_t = (u_1^{(t)}, \ldots, u_{N_z}^{(t)}) \in \{0,1\}^{N_z}$. A single PPO head is used.

5.4. Reward Function and PPO Architecture

The per-step reward is the negative interval cost minus a linear degree-hour comfort penalty:

$$r_t = -c^{(t)} - \alpha\Big[\max\big(0, T^{(t+1)} - T_{\max}\big) + \max\big(0, T_{\min} - T^{(t+1)}\big)\Big]\Delta t, \tag{20}$$

with $\alpha = 50\,\mathrm{SAR\ per\ ^\circ C\,h}$, and at the last step of an episode the terminal thermal state is priced exactly as the dynamic programme and the simulator price it. Under the rule layer the penalty is zero except on plant-limited cold drift. The policy and value networks are separate, each with two fully connected layers (256–128 units, tanh); the policy outputs the logits of a categorical distribution over $u \in \{0,1\}$. Training uses Stable-Baselines3 v2.9.2 [44] with Adam at a constant learning rate of 3×10^{-4}, $\gamma = 0.999$, clip coefficient 0.2, entropy coefficient 0, rollouts of 1,344 steps (two episodes) with minibatches of 336, and otherwise default settings; no hyperparameter search was made. Training runs for 2×10^5 environment steps (about 298 episodes of 672 steps, one 168 h window at $\Delta t = 15\,\mathrm{min}$) with five seeds (42–46), none discarded, a fixed budget and no checkpoint selection. Each episode starts at $T^{(0)} = 23\,^\circ\mathrm{C}$ on one of the six tuning windows of Section 7, so the policy is trained on six weekly weather sequences. During training the ceiling is the band edge, $\overline{T} = T_{\max}$, so R1 fires at 23.80 °C and R2 at 22.40 °C; in Algorithm 1, $\mathcal{R}(\cdot;\overline{T})$ applies R2 and R1 with $T_{\max}$ replaced by the ceiling $\overline{T}$, and evaluation takes the greedy action.

No convergence claim is made. With the rules on and a flat price, the training reward, which has no term in the mean temperature, is nearly minimised by holding the previous action until a rule fires (within 0.4, 1.4 and 2.3% of the exact $\lambda = 0$ dynamic-programming optimum inside the rules, called E1). Section 9.2 reports how close the trained seeds come to that policy and what longer training could gain. A second variant, E2, tests whether this is a property of the reward: it adds the stage term $(\lambda/n_{\mathrm{occ}})\max(T^{(t+1)}, T_{\min})$ of MPC, with λ taken from the in-rules 6 h MPC, and a perfect outdoor forecast at +1, +2, +3 and +6 h and irradiance at +1 and +3 h to the state; it is trained in Riyadh with three seeds, an entropy coefficient of 0.01 and reward normalisation, under the same rules in training and evaluation, with no Rule A selection. It stays close to the rules' hold policy (Section 9.2). Algorithm 1 sets out training and schedule extraction together; the policy π_θ it returns is the one evaluated in every RBRL row of Section 8.

5.5. Competing Methods

THERM (unoptimised baseline) models the automatic remote-controller thermostat common in Saudi residential buildings [19]: on when $T_i \geq T_{\max} - 0.5$, off when $T_i \leq T_{\min} + 0.5$, with no tariff awareness. On the occupied band adopted here this is a 1 °C deadband centred on 23 °C, assumed, not measured, to represent the split units in this stock (Section 9.2). Every optimising controller gains over THERM by using more of the 2 °C band under the same comfort obligation.

TUNED (rule-based comparator) is the same hysteresis family with its two switch points free. They are searched on a 0.05 °C grid and then refined iteratively on a 0.01 °C grid around the best admissible pair; a row at the 0.1 °C resolution of typical hardware is in the supplement. Without TUNED, any gap attributed to learning or planning may in fact be obtainable by commissioning the thermostat properly.

Guard hysteresis (the guard) is the hysteresis at the rule layer's own thresholds, R1 and R2, with no learned policy: on at $\overline{T} - \epsilon_{\mathrm{hot}}$, off at $T_{\min} + \epsilon_{\mathrm{cold}}$, with the ceiling $\overline{T}$ fitted like every other parameter. **D1** is a one-step model shield on a latch that holds the previous action and has no R1/R2 guard bands: it forces the unit on when the exact one-step prediction with the unit off would end above $\overline{T} - \sigma$, and off when the prediction with it on would end below $T_{\min} + \sigma$, with its own ceiling $\overline{T} \leq T_{\max}$ fitted for each margin $\sigma \in \{0, 0.05, 0.10\}\,\mathrm{K}$; on the trained PPO policies it replaces the rule layer. It needs the plant model and the current outdoor temperature and irradiance, where TUNED needs only the zone temperature. D1 separates what an exact one-step model buys from what a longer horizon buys.

MPC (causal receding-horizon comparator). At every step the controller solves, by dynamic programming over a temperature grid

Algorithm 1 RBRL: training, ceiling selection and evaluation

Require: simulator (4); tuning windows $\mathcal{W}_{\text{tune}}$ and held-out windows $\mathcal{W}_{\text{eval}}$ (start dates d_0, $N = 672$ steps each); price p; c_{sw}; budget $B = 2 \times 10^5$ steps; guard margins $\epsilon_{\text{hot}}, \epsilon_{\text{cold}}$; ceiling grid $\mathcal{C} = \{23.50, 23.51, \ldots, 24.00\}\,^\circ\text{C}$

Ensure: policy π_θ, ceiling $\overline{T}^*$, held-out costs $\{J_w\}_{w \in \mathcal{W}_{\text{eval}}}$

Phase 1: training ($\overline{T} = T_{\max}$)

Initialise π_θ; $n \leftarrow 0$
while $n < B$ **do** ▷ fixed budget; no early stopping
  Draw $d_0 \in \mathcal{W}_{\text{tune}}$; $T^{(0)} \leftarrow 23\,^\circ\text{C}$, $u^{(-1)} \leftarrow 0$
  for $t = 0$ **to** $N - 1$ **do**
    observe s_t (16); $a_t \sim \pi_\theta(\cdot \mid s_t)$
    $u^{(t)} \leftarrow \mathcal{R}(a_t, T^{(t)}; \overline{T})$ ▷ R1–R2, eq. (19)
    step (4); r_t by (20); $n \leftarrow n + 1$
  end for
  add the terminal price of $T^{(N)}$ to r_{N-1}
  every 1,344 steps: PPO update (clipped surrogate, GAE)
end while

Phase 2: R1-ceiling selection on the tuning windows (Rule A)

for $\overline{T} \in \mathcal{C}$ **do**
  run $\arg\max_a \pi_\theta$ with $\mathcal{R}(\cdot; \overline{T})$ on every $w \in \mathcal{W}_{\text{tune}}$
end for
$\overline{T}^* \leftarrow$ admissible $\overline{T}$ (Rule A) of least equalised tuning cost ratio

Phase 3: held-out evaluation (frozen π_θ, $\overline{T}^*$)

for $w \in \mathcal{W}_{\text{eval}}$ **do**
  run $\arg\max_a \pi_\theta$ with $\mathcal{R}(\cdot; \overline{T}^*)$; record J_w and $\bar{\tilde{T}}_w$
end for
return π_θ, $\overline{T}^*$, $\{J_w\}$ ▷ compared by Rule C, Section 7

with the first step taken from the exact state, the horizon problem with stage cost $c^{(t)} + (\lambda/n_{\text{occ}}) \max(T^{(t+1)}, T_{\min})$ over occupied steps, applies the first action and re-solves. Cooling steps that would end below $T_{\min}$ are infeasible, as for every optimiser here; if no action had a finite planned cost, the TUNED action would be applied, which never occurred. The multiplier λ is the comfort device: one value per city, horizon and forecast mode, fitted on the tuning windows by Rule A of Section 7 and then frozen. If no multiplier is admissible even at the 0.10 K fallback, MPC's occupied ceiling is lowered by a back-off b fitted by the same rule (cells marked †); the order is λ at 0.03 K, then λ at 0.10 K, then b. For the 6 and 24 h perfect-forecast controllers the fitted values lie within 2.5% of $\lambda^* = \Lambda_i\,(168\,\text{h})\,p/\text{COP}$, the marginal envelope cost of 1 K over a window, at which the Lagrangian is indifferent to the temperature level; the controller therefore minimises on-commands and its comfort level is set by the Rule A selection of λ. Close to λ^* the result is sensitive to λ: in Riyadh, neighbouring admissible multipliers give 24 h savings that differ by 1.4–1.8 points at both grid resolutions, so the fitted λ/λ^* is reported in Section 8.3. The horizon problem is closed by a rollout terminal cost: the cost, including the on-command charge and the λ term, of running TUNED for 24 h from the terminal state on a persistence forecast, plus the price of the stored heat. This is a truncated rollout of a base policy [45]; the classical guarantee of closed-loop cost no greater than the base policy's does not carry over, because the tail is truncated, uses persistence weather and is evaluated on a grid, so any improvement over TUNED is an empirical finding. Horizons of 0.25, 1, 2, 6, 24 and 48 h are run with two forecast modes: *perfect*, the true in-horizon weather, and *persistence*, in which the forecast for step j is the observation one day earlier, with the current outdoor temperature measured. In both modes the controller plans on the plant's own model with the exact state. The *in-rules* variant restricts every action to the training rule set, R1/R2 at 23.80/22.40 °C; its terminal rollout still uses TUNED, whose off point lies below R2. The value grid has 1,601 nodes; at the fitted λ, refining it to 3,201 nodes changes the 24 h saving by 0.10, 0.45 and 0.29 points in Riyadh, Jeddah and Taif; the grid uncertainty is taken as at most 0.5 points, below the 3-point resolution rule.

CLV (clairvoyant reference) is a dynamic programme over the exact dynamics (3) that sees the whole window's weather. Its multiplier is set per held-out window, by bisection (with an occupied-ceiling cap as a second knob where needed), aiming at a clipped mean within 0.002 K of THERM's. At the critical multiplier the optimal schedule's mean jumps by 0.1–0.3 K, so the target is met in only 2 of 7 windows in Riyadh, 1 of 7 in Jeddah and none in Taif; elsewhere the closest schedule no warmer than THERM is kept, and its residual is recorded (per window in the supplement).

End-of-window convention. The simulator, and

so every causal arm, prices the terminal thermal state at the electricity needed to return it to the initial temperature T_0 at rated COP. A programme that knows where the window ends could exploit that price, because it omits the on-command the return would need. The oracles (CLV, the bound and B_a) therefore use $V_N = \big[p\,C_{\mathrm{th}}/(3.6\times10^6\,\mathrm{COP}) + c_{\mathrm{sw}}/(1\,\mathrm{K})\big](T_N - T_0)$: heat left above T_0 is charged its re-cooling energy plus the fraction $(T_N - T_0)/1\,\mathrm{K}$ of one on-command, that is, one on-command per kelvin, which is a convention. The savings of the causal arms in Table 3 use the simulator terminal, the bracket and B_a use V_N, and B^* is also reported under the simulator terminal for Riyadh and Jeddah. E1 keeps the simulator terminal, which is PPO's own objective. CLV and the bound of Section 8.1 bracket what a schedule no warmer than the baseline could save; the gap between TUNED and that bracket is the headroom any controller could in principle occupy.

Genetic-algorithm and simulated-annealing schedulers [35, 46] are not run. Both are day-ahead offline solvers, and the bound of (23) applies to them as to every other schedule at the baseline's mean, so it answers the question they would be introduced to answer. No third-party learned comparator (SAC, DQN) is run inside the rule layer either. A relaxed bound restricted to the training rule set (14.9, 15.6 and 16.2% in Riyadh, Jeddah and Taif) caps what any such learner could attain, but whether another learner would come closer to it than PPO is untested, as is any learner acting outside R1–R2.

The framework is applied to the three climates characterised next.

6. Case Studies: Riyadh, Jeddah and Taif

Three sites are used, chosen to span the conditions under which a Saudi dwelling is cooled, not to represent the Kingdom: hot-arid inland (Riyadh), hot-humid coastal (Jeddah) and mild highland (Taif).

6.1. Climate Profiles

Riyadh (BWh, 24.7°N, 635 m) is an inland desert city with summer highs reaching 46.6 °C and a cooling-season diurnal swing of 12.7 K, the largest of the three sites (CDD_{18} = 3,693; cf. [47]). Jeddah (BWh, 21.7°N, 15 m) is a Red Sea coastal city, classified hot-humid [48]; in the record used here its annual mean relative humidity is 58.5% with 17.5% of hours above 70%, and it has a compressed diurnal swing of 8.4 K and little overnight relief (CDD_{18} = 4,260; the ordering of the two totals depends on the typical-year record, and what distinguishes the sites here is the diurnal swing). Taif (BSh, 21.5°N, 1,478 m) is a highland site whose altitude lowers both the peak and the mean: over the cooling season its mean outdoor temperature is 28.6 °C against Riyadh's 35.0 and Jeddah's 32.7; its peak 39.4 °C against 46.6 and 44.0; and its CDD_{18} = 2,277 against 3,693 and 4,260. Its diurnal swing, 11.7 K, is close to Riyadh's. Its load is the smallest of the three, so it leaves the least to schedule in absolute terms, although its headroom in percentage terms is not the smallest (Section 8.2). The lower air density at that altitude is also why the infiltration conductance is evaluated per city (Section 6.2). Monthly and diurnal profiles are in the repository. All three simulations use TMYx typical-year records from `climate.onebuilding.org`, WMO 404380 (Riyadh, period of record 2009–2022, 12 years of data), 410240 (Jeddah, 2009–2023, 15 years) and 410360 (Taif, 2009–2023, 15 years); hourly dry-bulb temperature and the direct-normal and diffuse-horizontal irradiances are read from fields 7, 15 and 16, and the glazing gain is the plane-of-array irradiance on a south-facing vertical surface under an isotropic sky with ground albedo 0.30.

6.2. Building Code, Tariff, Humidity, and Comfort Standard

SBC 2018 mandates opaque-wall U-value $\leq 0.45\,\mathrm{W\,m^{-2}K^{-1}}$ and window SHGC ≤ 0.25 in both Climate Zone A (Jeddah) and Zone B (Riyadh) [16]. Table 2 meets both at assembly level: $U_{\mathrm{wall}} = 0.45$ and SHGC = 0.25 exactly, with $U_{\mathrm{win}} = 2.8\,\mathrm{W\,m^{-2}K^{-1}}$ consistent with double glazing. The step-wise tariff follows equation (11); for Models L and E, $c_r^{\mathrm{L}} = 0.12\,\mathrm{SAR/kWh}$ is adopted as an illustrative flat rate, below the post-2018 first-block price of 0.18 SAR/kWh. The occupied comfort band $[22, 24]\,^\circ\mathrm{C}$ is consistent with the *thermal* criterion of ASHRAE 55-2023 [49] for mechanically conditioned residential spaces (the standard's humidity criterion is not evaluated here; Section 10.4). Bienvenido-Huertas et al. [50] validate this use, and its upper limit reflects Saudi practice: Krarti [51] recommends 23–25 °C for Kingdom dwellings and observes setpoints as low as 18 °C in the stock. Lamsal et al. [52] review adaptive thermal comfort models and find wider deadbands appropriate

for energy-efficient design in hot climates; the narrower band adopted here raises the reported saving through the baseline's cycling (Section 10.3). A fixed band is adopted because Saudi residential buildings are predominantly mechanically cooled; ASHRAE 55 Section 5.3 restricts adaptive comfort to naturally ventilated spaces. The saving does not transfer across band widths, because the actuation condition (8) and the baseline's cycling rate both depend on the width.

7. Simulation Setup

The controllers of Section 5 are evaluated on the sites of Section 6 under one protocol, set out here. All simulations run in Python 3.13.18 with NumPy 2.5.3, integrating equation (3) exactly at $\Delta t = 15\,\text{min}$, the interval that satisfies (8) with a factor of 2.7 (Remark 1). Solar gain is $Q_{\text{sol}}^{(i,t)} = \text{SHGC} \times A_{gl} \times I^{(t)}$, with $I^{(t)}$ the plane-of-array irradiance of Section 6.1; hourly weather values are held constant over the intervals of each hour, without interpolation. The RC model has no occupancy or internal gains. In the headline and every sensitivity every step is occupied; only the setback arms of Section 8.2 use a timetable.

Background load and tariff.. The tariff is levied on the whole account, so the cumulative energy E_{cum} starts from a non-HVAC load of $E_{\text{base}} \approx 800\,\text{kWh/month}$ [2]. The single zone adds about 84 kWh a month, so the account reaches ≈884 kWh and stays well below the 6,000 kWh boundary (Remark 3).

Evaluation protocol.. Evaluation is confined to the cooling season, 1 May to 31 October, because the plant is cooling-only. Thirteen windows of 168 h start every 14 days, at zero-based day indices 121 to 289 (1 January is day 0), none overlapping, and are assigned alternately within one TMYx year. The seven *held-out* windows start on 2 May, 30 May, 27 June, 25 July, 22 August, 19 September and 17 October; the six *tuning* windows start on 16 May, 13 June, 11 July, 8 August, 5 September and 3 October. There is no separate validation set: every parameter, including PPO's ceiling, is selected on the tuning windows alone, PPO trains for a fixed budget with no early stopping or checkpoint selection, and the held-out windows are touched only once, for evaluation. The seven held-out windows span the cooling season, so the seasonal variation is carried by the per-window tests rather than by a separate seasonal evaluation; the inter-annual spread is measured on actual years (Section 10.2). One protocol, implemented in one code module, applies to every arm and oracle.

Comfort. On every occupied step the post-step temperature must not exceed $T_{\max}$; the occupied over-temperature in degree-hours is recorded for every arm. On the cold side, *cooling may never end a step below $T_{\min}$; free-float is unrestricted* (Definition 6), so sub-band states arise only by free drift when the off-state equilibrium lies below the band. In the optimisers this is a hard constraint on cooling transitions. The rule-layer arms and D1 are designed to meet it from the nominal plant parameters, and TUNED meets it on the tuning windows by Rule A and threshold placement, not by guarantee; for all of them it is measured as commanded under-temperature, the degree-hours below $T_{\min}$ at the end of cooling steps, reported beside the free-float under-temperature of idle occupied steps.

Rule A (selection on the tuning windows). Each causal arm has one parameter set per city: TUNED's two switch points, the guard's and RBRL's ceiling, D1's margin σ, and MPC's λ for each horizon and forecast mode. A candidate is admissible if, on every tuning window, it has zero over-temperature and zero commanded under-temperature, and if its clipped mean occupied temperature pooled over the tuning windows lies within 0.03 K of THERM's. Among admissible candidates the one with the least equalised tuning cost ratio $\sum_w J_w / \sum_w J_{\text{THERM}(\delta_w),w}$ (Rule C applied to the tuning windows) is selected and frozen. If no candidate meets the 0.03 K condition, it is relaxed to 0.10 K, for every arm alike; each table marks which applies, and an arm with no admissible parameter is reported as such and not tabulated. A sensitivity refit under a per-window 0.03 K criterion is in the supplement.

Rule B (clipped mean). Occupied means are computed on $\tilde{T} = \max(T, T_{\min})$ for every arm and for THERM, so a free-float cold excursion buys no credit against warmth elsewhere.

Rule C (equalisation per held-out window). On each held-out window w the frozen arm is compared with THERM shifted by δ_w (on at $23.5 + \delta_w$, off at $22.5 + \delta_w$), with δ_w scanned over $[-0.4, 0.4]\,\text{K}$ in 0.002 K steps. THERM's clipped mean is not monotone in δ, so the arm's mean can be reached at

several shifts; the comparator cost is the *minimum* over all crossings of the piecewise-linear scan, interpolated linearly within each bracketing segment, which credits the baseline with its cheapest setting of equal comfort and can only understate a saving. Scored this way, the unshifted thermostat itself saves −0.8, −0.1 and 0.0% in Riyadh, Jeddah and Taif, which is the size of the rule's conservatism; the range between the cheapest and the dearest crossing is in the supplement. The only saving printed is the pooled

$$s = 1 - \frac{\sum_w J_{\mathrm{arm},w}}{\sum_w J_{\mathrm{THERM}(\delta_w),w}}, \tag{21}$$

over the seven held-out windows; the raw saving against unshifted THERM, the pooled and largest per-window mean differences, the comfort metrics and the pooled end temperature are printed beside it. Per-window savings enter only the tests. No held-out window contributes to any arm's parameters. CLV is matched per held-out window (Section 5.5), and the bound needs no selection (Section 8.1).

Statistics. The unit is the per-window difference in equalised saving between two arms on the same seven windows. Each contrast is tested with the exact two-sided Wilcoxon signed-rank and sign tests and summarised by the Hodges–Lehmann estimate with an exact 95.3% interval (the third smallest and largest of the 28 Walsh averages); the estimate and interval lead, the p-value is secondary. The primary families, each one contrast in three cities with Holm's correction within the family (error control is per family), are F1, TUNED against THERM; F2, MPC (24 h, perfect) against TUNED; F3, MPC (24 h, persistence) against TUNED; and F4, the guard against TUNED; E2 is not tested against the guard, because its comfort is not matched to the other arms. With $n = 7$ and three cities, 0.047 is the smallest Holm-adjusted p attainable, reached by a city whose seven windows agree in sign (or, for the Wilcoxon test, with one discordant window of small rank when the other two cities are unanimous); results short of it are reported as counts. The horizon, capacity, switching-charge, deadband and tariff sweeps are descriptive. Pooled differences of less than 3 points between arms are treated as not resolved, regardless of the test result, because the occupied-mean staircase and the tuning noise are of that order; the threshold does not absorb comfort offsets between arms (Section 10.4). The held-out weeks come from one TMYx year, so the tests describe those weeks. Estimates, intervals, p-values and comfort offsets for every family are in Supplementary Table S7.

Switching charge. For a split unit of ∼2,500 SAR installed cost, a start life of order 10^5 gives 0.025 SAR per start, and a 15-year life at about 1,460 on-commands a year (four a day all year) gives about 0.114 SAR; counting only the 184-day cooling season at THERM's 4.10 a day would give about 0.22 SAR. The range 0.025–0.114 SAR per on-command is therefore a convention, not a measured wear cost; none was found for Saudi split units [53]. The tables use $c_{\mathrm{sw}} = 0.15$ SAR as the computational reference, because the PPO policies were trained at it; it lies above the derived range and so overstates the saving relative to that range, because the saving rises with c_{sw}. Results at 0.025 and 0.114 SAR are reported beside it (Section 9.2). Every percentage saving is relative to the room's cooling energy cost plus on-command charges, not to the household bill.

8. Results

Results are reported on the seven held-out windows at the tariff in force, $c_{\mathrm{sw}} = 0.15$ SAR, $\Delta t = 15$ min and $C_{\mathrm{th}} = 4{,}800$ kJ/K, with MPC on a 1,601-node grid. Every fitted single-zone arm records zero occupied over-temperature on the held-out windows; the two fixed setback arms do not (Section 8.2). Commanded under-temperature (Supplementary Table S4) is zero for every optimiser, rule-layer and D1 arm except the Riyadh 6 h persistence controller, whose plan meets the cold rule on the forecast but not on the realised weather (0.0006 °C·h over the seven windows). The threshold arms meet the cold rule only on the tuning windows: on the held-out windows TUNED records 0.0007 and 0.0066 °C·h in Jeddah and Taif, and TUNED-SB 0.0102 °C·h in Riyadh, the largest in the headline configuration (0.024 °C·h at most in any sensitivity run). The largest free-float value, 0.96 °C·h, belongs to the Taif 24 h persistence controller, and in Taif TUNED and the guard also free-float below the band (0.29 and 0.34 °C·h). The multi-zone controller's record is given in Section 8.4. Section 8.1 first fixes what clairvoyance is worth; Sections 8.2–8.4 then read the controllers against it.

8.1. *What Clairvoyance Is Worth: a Bound and an Attained Schedule*

The bound keeps comfort pointwise and applies the comfort match only in the mean, so it covers every schedule no warmer than the baseline:

$$\begin{aligned} \min_{u}\ & J(u) \quad \text{s.t.} \quad T^{(t+1)} \le T_t^{\mathrm{hi}} \ \ \forall t, \\ & u_t = 1 \ \Rightarrow\ T^{(t+1)} \ge T_{\min} \ \ \forall t, \\ & \tfrac{1}{|\mathcal{O}|}\textstyle\sum_{t\in\mathcal{O}} \tilde{T}^{(t+1)} \ \le\ T^{\star}, \end{aligned} \tag{22}$$

with $\mathcal{O}$ the occupied steps, $\tilde{T} = \max(T, T_{\min})$, $T_t^{\mathrm{hi}} = T_{\max}$ while occupied and the empty-house limit otherwise, and $T^{\star}$ the baseline's clipped mean. As in the protocol, a free-floating step may end below the band and the clipped mean gives it no credit (Rule B). The mean constraint is an inequality because only warmth can buy a saving, and because with binary actions on a 15 min grid the attainable means form a finite set. Relaxing it alone gives $L(u,\lambda) = J(u) + \lambda\big(|\mathcal{O}|^{-1}\sum_{t\in\mathcal{O}} \tilde{T}^{(t+1)} - T^{\star}\big)$, the same dynamic programme with one additive stage cost, and for any $\lambda \ge 0$ and any u feasible for (22)

$$J(u) \ \ge\ L(u,\lambda) \ \ge\ \min_{u'} L(u',\lambda) \ =\ L^{\star}(\lambda). \tag{23}$$

The attained mean is a step function of λ, so the dual gap need not close and no exactly mean-matched schedule need exist in the Lagrangian family.

A provable bound on a grid. $L^{\star}(\lambda)$ is computed by dynamic programming on a uniform temperature grid of spacing h, and an ordinary discretised programme can return a minimum that is too *expensive*, the unsafe direction for a bound. The reported bound therefore uses an optimistic-neighbour relaxation. For each node y, previous action and action u, the exact successor $f_u(y)$ is computed; energy and on-command costs are charged exactly; the mean term is charged at $\max(f_u(y) - h, T_{\min})$; feasibility is tested against limits widened by h; the continuation is the smaller of the values at the two nodes bracketing $f_u(y)$; the terminal value is evaluated at $y - h$; and the initial value is the smaller of the values at the two nodes bracketing $T^{(0)}$. The grid covers the band plus 0.1 K and extends below it to the window's lowest free-float equilibrium, so no trajectory admitted by (22) leaves it.

Proof. Take any feasible u with true trajectory x_t. Let y_0 be the node bracketing x_0 nearest to it, and y_{t+1} the node bracketing $f_{u_t}(y_t)$ nearest to x_{t+1}. The plant map $f(x) = T_\infty + (x - T_\infty)a$, $a = e^{-\Delta t \Lambda / C_{\mathrm{th}}} \in (0,1)$, gives $|f(y_t) - x_{t+1}| = a|y_t - x_t|$, so $|y_t - x_t| \le h$ implies $|y_{t+1} - x_{t+1}| \le h$, and by induction this holds for every t. Along this path each relaxed stage cost is at most the true one: the mean term because $f(y_t) - h \le f(x_t)$ and $\lambda \ge 0$; the terminal value because it increases with T and $y_N - h \le x_N$. Every true-feasible step is relaxed-feasible, because $|f(y_t) - f(x_t)| \le h$ (and, for the rule-constrained variant, whose masks test the current state, because $|y_t - x_t| \le h$). Because the relaxed continuation takes the smaller value of the bracketing pair, the relaxed value at y_0 is no larger than the relaxed cost along this path. Hence $L_h^{\star}(\lambda) \le J(u) + \lambda\big(\bar{\tilde{T}}(u) - T^{\star}\big) \le J(u)$ for every $\lambda \ge 0$ and every u feasible for (22), pooled over the held-out windows. $L_h^{\star}$ is concave in λ, so λ is maximised on an eighteen-point grid with ternary-search refinement; an under-maximised λ weakens the bound without invalidating it. □

The relaxation has a price. The bracketing minimum may round the successor down by up to h at every step, and the roundings accumulate over the plant's memory, $1/(1-a) \approx 225$ steps, so the slack is of order $h/(1-a)$ in temperature and shrinks roughly in proportion to h: under V_N the relaxed saving bound is 21.3, 20.6, 20.2 and 20.0% in Riyadh at 1,601, 3,201, 6,401 and 12,801 nodes (21.2, 20.5, 20.1 and 19.8% in Jeddah), against 19.5% and 19.1–19.4% from the ordinary programme, which is not a bound; at 12,801 nodes the slack is 0.47 and 0.42 points. In Taif the relaxed bound is 23.8, 22.2, 21.4 and 20.9% against 19.9–20.2%, a slack of 0.75 points at 12,801 nodes. Every grid gives a valid bound, so the smallest is reported, rounded up to 0.1. The bound is a minimum over every action sequence in the window under the same cost, so it holds for any controller whatever information it uses; MPC's forecast beyond the window end cannot take it above the bound, because post-window weather enters the cost only through the terminal price of the stored heat.

What is reported. The headline bound is $B^* = 1 - L^{\star}(T^{\star})/\sum_w J_{\mathrm{THERM}(0),w}$ at THERM's pooled mean. Each arm a is warmer or cooler than THERM by its own $\Delta\tilde{T}$, so its bound is evaluated at its own mean, $B_a = 1 - L^{\star}(\bar{\tilde{T}}_a)/\sum_w J_{\mathrm{THERM}(\delta_w^a),w}$, and $s_a \le B_a$ then holds exactly once each arm is rescored under V_N; B_a for the principal arms is in Supplementary Table S5 (all 89 rows in the results

files). The lower end of the bracket is the *best attained schedule no warmer than THERM*, scored as $1-\sum_w J_{\mathrm{CLV},w}/\sum_w J_{\mathrm{THERM}(0),w}$, with the same denominator and the same mean constraint as B^*; the upper end is the provable bound under the model; the gap between them is the sum of the duality gap, the relaxation slack and the replay's own sub-optimality. CLV's equalised saving under Rule C is printed only beside B_{CLV}, the bound at CLV's own mean, never against B^*.

Here the bracket is [18.0, 20.0]% in Riyadh, [18.4, 19.9]% in Jeddah and [16.7, 21.0]% in Taif, 2.0, 1.5 and 4.3 points wide. Taif's relaxation slack (0.75 points) explains little of its width; the rest is duality gap and replay sub-optimality, and CLV meets the comfort target in none of Taif's windows. Under the simulator's energy-only terminal B^* is 20.0 and 20.1% in Riyadh and Jeddah (not computed for Taif). CLV's equalised saving is 18.3% against $B_{\mathrm{CLV}} = 19.6\%$ (rounded to nearest, as is B_a) in Riyadh, 18.5% against 19.4% in Jeddah and 16.5% against 20.2% in Taif; end-of-window temperatures are in Supplementary Table S5. Every arm satisfies $s_a \le B_a$ and every shifted per-window certificate is non-negative under V_N, in all 89 arm and seed rows checked (every causal arm, E2, the D1 shield, MPC in both rule sets, and the setback arms against the setback bound; the rule-layer arms also against the ruled bound). It is a bracket at the instance evaluated (the tariff in force, $c_{\mathrm{sw}} = 0.15\,\mathrm{SAR}$, $\delta_{\mathrm{db}} = 1\,^\circ\mathrm{C}$, $\Delta t = 15\,\mathrm{min}$), not at other deadbands or tariffs.

8.2. Cost, Energy and On-Commands

Table 3 gives every arm under the protocol of Section 7, and Table 4 reads the principal arms in three layers. Each saving is the pooled equalised saving (21), which removes the first-order energy effect of the mean temperature; a residual dependence remains (comfort-offset caveat, Section 10.4).

Three layers. Every result is read in three layers (Table 4). The physical headline is the share of on-commands avoided against the comfort-equalised thermostat, which needs no price: 22.8, 24.2 and 21.7% for TUNED, 29.8–33.3% for the 24 h MPC with a perfect forecast, and 29.2–42.3% with occupancy setback; Fig. 3 shows where they come from. Beside it, electricity is about unchanged: within ±1.2% for every arm without setback and up to +3.3% for MPC with setback, which cools the empty house before arrival. At equal mean temperature every arm uses the same terminal-priced energy (the identity holds to within 4×10^{-5} relative). Priced at the derived 0.025–0.114 SAR per on-command, TUNED is worth 4–19, 5–19 and 2–10 SAR per room per cooling season, and the 24 h MPC 6–28, 6–26 and 4–15 SAR (Table 4 adds the figures at the 0.15 SAR reference, which overstates), so the cost difference is the on-command charge (Remark 2). The pooled cost saving s remains the protocol's ranking metric (Table 3).

The tuned thermostat. Commissioning the two switch points saves 12.5, 13.6 and 12.4% of cost in Riyadh, Jeddah and Taif, in every held-out window of every city (family F1: Hodges–Lehmann +12.6, +13.7 and +11.8 points, Holm-adjusted $p = 0.047$ in each city), with no model, no forecast and no state beyond the zone temperature. The mechanism is a wider deadband: the fitted switch points (Section 5.3) widen THERM's 1 °C deadband to 1.35, 1.36 and 1.32 °C, above the 1.25 °C limit for a centred deadband but within the off-centre upper limit of (8) (Remark 1; TUNED is admitted empirically by Rule A), and the share of on-commands avoided is roughly $1 - \delta_{\mathrm{THERM}}/\delta_{\mathrm{TUNED}}$. The saving is therefore a statement about the comparator's assumed 1 °C deadband as much as about tuning (Section 9.2). At the charges the wear derivation supports, 0.025 and 0.114 SAR, it saves 3.8–11.0%, 4.3–12.0% and 3.9–10.9%. These switch points are fitted on a 0.01 °C grid. A real thermostat sets its thresholds in steps of about 0.1 °C, and on that grid the tuned rule avoids 10.5, 13.2 and 14.7% of on-commands and saves 5.7, 7.4 and 8.4% (at 23.6/22.4 °C in every city, which in Riyadh is the guard hysteresis, so on that grid the guard's shortfall F4 vanishes in Riyadh, and in Taif the guard saves more than TUNED, by less than 3 points, not resolved); under a per-window 0.03 K admission criterion it saves 9.7, 12.2 and 4.0%, 8.4 points less in Taif, where the guard then exceeds it (Supplementary Table S2). The headline TUNED figures are therefore an upper estimate of what a commissioned thermostat delivers. Any claim for a learned or model-based controller is measured against TUNED, not THERM.

Taif. The lower load does not leave less to win in percentage terms: TUNED saves 12.4% against a bracket of 16.7–21.0%, the widest of the three, with the lowest lower end and the highest upper end. The 24 h MPC's 16.9% uses the simulator terminal; under V_N its equalised saving is 16.3%, below CLV's 16.5%.

Under the schedule announced for January 2027

Table 3: Held-out results at the tariff in force, occupied band $[22, 24]\,^{\circ}\mathrm{C}$, $\Delta t = 15\,\mathrm{min}$, $c_{\mathrm{sw}} = 0.15\,\mathrm{SAR}$, single zone with one exposed face. s: pooled equalised saving (21) (%); $\Delta\tilde{T}$: pooled clipped-mean difference from THERM (K); on: change in on-commands against the equalised baseline (%). Fit status: ^{a}pooled 0.03 K; ^{b}pooled fallback 0.10 K; ^{o}oracle. Rows whose $\Delta\tilde{T}$ differs by more than about 0.02 K carry the comfort-offset caveat (about 1.3 points per 0.01 K along the guard family, less along the MPC family; Section 10.4). MPC rows: 1,601 grid nodes, perfect (p) or persistence (s) forecast; †comfort matched by occupied-ceiling back-off b, because the λ device alone cannot reach the matched comfort level. "training rules": R1/R2 at 23.80/22.40 °C; "guard rules": each city's fitted guard thresholds. D1 rows at $\sigma = 0$. CLV: *raw* saving against unshifted THERM, the best attained schedule no warmer than THERM; its Rule C saving is paired with B_{CLV} in the text. Causal arms use the simulator's energy-only terminal; the CLV and bound rows use V_N (Section 5.5); B^* is given under both.

	Riyadh			Jeddah			Taif		
Arm	s	$\Delta\tilde{T}$	on	s	$\Delta\tilde{T}$	on	s	$\Delta\tilde{T}$	on
THERM (J, on/day)	1.118, 4.10			0.993, 3.73			0.614, 2.33		
TUNED	12.5^{a}	+0.011	−22.8	13.6^{a}	+0.029	−24.2	12.4^{a}	+0.005	−21.7
Guard hysteresis	5.7^{a}	+0.036	−10.5	7.6^{a}	+0.035	−13.6	11.0^{a}	+0.056	−19.3
RBRL, 5 seeds	5.7^{a}	+0.036	−10.5	7.6^{a}	+0.035	−13.6	8.6–11.0^{a}	—	—
D1 shield on latch	14.0^{a}	−0.009	−25.4	13.5^{a}	+0.021	−24.0	12.4^{a}	+0.005	−21.7
D1 on PPO, 5 seeds	12.8–14.1^{a}	—	—	11.2–13.5^{a}	—	—	8.0–12.0^{a}	—	—
MPC 0.25 h, p	−3.9^{a}	+0.058	+7.1	1.6^{a†}	+0.023	−2.9	1.0^{a}	+0.034	−1.8
MPC 1 h, p	10.1^{a†}	+0.024	−18.4	10.1^{a†}	+0.001	−17.9	6.8^{a†}	+0.009	−11.9
MPC 2 h, p	7.0^{b}	−0.099	−12.7	10.2^{a}	−0.062	−18.2	9.2^{a†}	−0.003	−16.2
MPC 6 h, p	15.1^{a}	−0.021	−27.4	16.3^{a}	+0.022	−28.9	15.9^{a}	−0.005	−28.0
MPC 6 h, s	14.4^{a}	−0.004	−26.3	no admissible parameter			14.0^{a}	+0.003	−24.6
MPC 24 h, p	18.3^{a}	+0.014	−33.3	18.1^{a}	−0.011	−32.0	16.9^{a}	−0.011	−29.8
MPC 24 h, s	12.7^{b}	−0.076	−23.2	14.6^{a}	−0.037	−25.9	10.8^{b}	−0.086	−18.8
MPC 48 h, p	17.6^{a}	−0.018	−32.0	17.5^{a}	−0.012	−31.1	18.9^{b}	+0.089	−33.4
MPC 1 h, training rules	9.7^{b}	+0.115	−17.7	−3.6^{a}	−0.053	+6.5	−1.8^{b}	−0.023	+3.1
MPC 6 h, training rules	10.7^{a}	−0.028	−19.4	10.9^{a}	−0.006	−19.3	11.1^{b}	−0.016	−19.4
MPC 24 h, training rules	11.6^{b}	−0.061	−21.1	14.1^{a}	−0.003	−24.9	10.0^{b}	−0.064	−17.6
MPC 1 h, guard rules	2.5^{a}	+0.038	−4.5	1.5^{b}	+0.051	−2.8	8.0^{b}	+0.064	−14.2
MPC 6 h, guard rules	6.4^{a}	+0.025	−11.7	7.6^{a}	+0.024	−13.5	9.5^{b}	+0.076	−16.7
MPC 24 h, guard rules	6.7^{a}	+0.028	−12.3	4.4^{b}	+0.068	−7.7	10.5^{a}	+0.043	−18.6
CLV, raw^{o}	18.0	−0.036	−31.1	18.4	−0.018	−31.2	16.7	−0.025	−26.1
B^*, energy-only / V_N	≤ 20.0 / ≤ 20.0			≤ 20.1 / ≤ 19.9			— / ≤ 21.0		
CLV in rules^{o} / ruled bound	12.9 / ≤ 14.9			13.8 / ≤ 15.6			12.3 / ≤ 16.2		

Table 4: The three layers of every result, held-out windows, tariff in force. On: on-commands avoided against the comfort-equalised thermostat (%), the physical headline. Energy: electricity saved (%; negative means more), about unchanged for every arm. SAR: saving per room per cooling season, $184\,[c_{\mathrm{sw}}(\bar{f}_{\mathrm{eq}} - \bar{f}_{\mathrm{arm}}) + p(\bar{E}_{\mathrm{eq}} - \bar{E}_{\mathrm{arm}})]$ with $\bar{f}$ the mean on-commands per day and $\bar{E}$ the mean electricity per day over the seven held-out windows, $p = 0.18\,\mathrm{SAR/kWh}$, at $c_{\mathrm{sw}} = 0.025$–$0.114\,\mathrm{SAR}$ (the derived range) and, in brackets, at the $0.15\,\mathrm{SAR}$ computational reference, which overstates. Setback arms are scored on the occupancy timetable as scheduled, against THERM without setback equalised on the occupied steps.

	Riyadh			Jeddah			Taif		
Arm	On	Energy	SAR/season	On	Energy	SAR/season	On	Energy	SAR/season
TUNED	22.8	−0.3	4–19 (25)	24.2	+0.5	5–19 (25)	21.7	−0.5	2–10 (14)
Guard hysteresis	10.5	−0.4	2–8 (11)	13.6	0.0	2–10 (14)	19.3	0.0	2–9 (12)
D1 shield on latch	25.4	+0.7	5–22 (29)	24.0	−0.7	3–18 (24)	21.7	0.0	2–11 (14)
MPC 24 h, perfect	33.3	−0.4	6–28 (37)	32.0	+1.2	6–26 (34)	29.8	+0.8	4–15 (20)
MPC 24 h, persistence	23.2	−0.2	4–20 (26)	25.9	+0.7	5–21 (27)	18.8	−1.2	1–9 (12)
TUNED-SB (setback)	33.6	−0.2	6–28 (37)	35.0	+0.6	6–28 (36)	29.2	−0.8	3–14 (18)
MPC-SB (setback)	42.3	+0.6	9–37 (49)	39.7	+2.1	8–33 (42)	39.3	+3.3	6–21 (28)

(0.21 SAR/kWh) TUNED saves 11.6, 12.7 and 11.6% while avoiding the same on-commands: a higher energy price dilutes a saving whose mechanism is a fixed charge per on-command.

Occupancy setback. With the dwelling empty from 08:00 to 14:00 on working days and an away limit of 30 °C, a tuned setback rule (TUNED-SB, with a fitted pre-arrival lead of 2, 1.25 and 1 h)

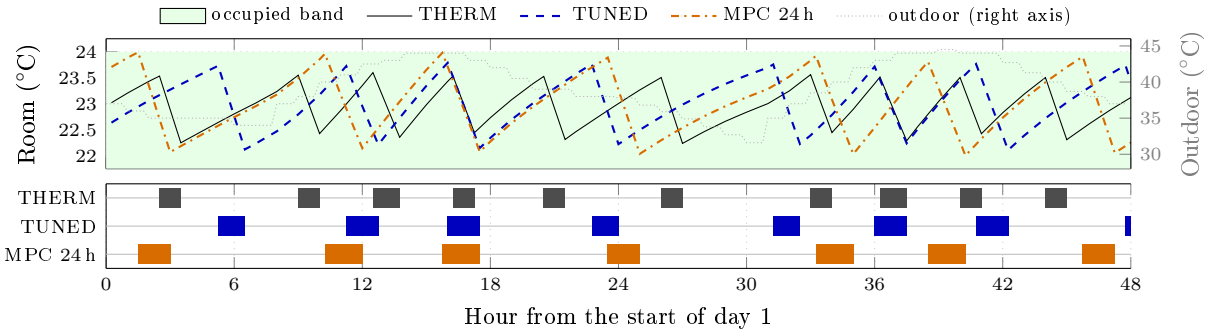


Figure 3: A two-day operating trace: the held-out Riyadh week with the highest mean outdoor temperature, days 1–2, fixed before the traces were drawn (day 0 is not shown). It is the held-out week in which TUNED's saving is lowest (11.1%); the MPC margin over it there is 6.3 points. Upper: room temperature at the end of each 15-min step under THERM (unshifted), TUNED and the 24 h MPC (perfect forecast), with the occupied band $[22, 24]$ °C shaded and the outdoor temperature (31.6–44.5 °C) on the right axis. Lower: cooling command (filled: on). THERM, TUNED and MPC issue 10, 8 and 7 on-commands over the two days and 34, 27 and 23 over the week. TUNED's switch points (23.71/22.36 °C) widen THERM's 1 °C deadband, so it cycles less; MPC uses the full band (22.00–23.99 °C).

saves 18.6, 20.1 and 17.0% ($\Delta\tilde{T} = +0.058, +0.027, +0.001$ K), and a 24 h MPC with the timetable (MPC-SB) saves 24.0, 23.3 and 23.2% ($\Delta\tilde{T} = -0.030, -0.019, -0.076$ K), against setback bounds of 27.8, 27.6 and 28.8% and raw setback clairvoyant schedules of 25.3, 25.3 and 23.9%. Both are compared with THERM without setback, equalised on the occupied steps, so the saving includes letting the empty house float. The pre-arrival lead is not pre-cooling: it is the latest start that meets the 24 °C ceiling at arrival, and no cooling is banked beyond it.

Switching off at departure and on at arrival with no lead (MANUAL) is a reference, not an admitted arm: it leaves 21.8, 9.8 and 3.3 occupied degree-hours above the band after arrival, and its nominal saving (14.6, 13.0 and 13.7%) is bought with that over-temperature; THERM with the setback away limit coincides with it, because the away limit never binds (the empty zone peaks at 24.8–25.9 °C). The mechanism is on-commands avoided while the house is empty: TUNED-SB avoids 33.6, 35.0 and 29.2% of the equalised baseline's on-commands while energy moves by −0.2, +0.6 and −0.8%, and MPC-SB avoids 42.3, 39.7 and 39.3% with energy +0.6, +2.1 and +3.3% (vacation weeks: Supplementary Section S8). MPC-SB plans on the timetable it is given, and the model has no internal gains and a constant capacity, both of which flatter the recovery.

8.3. Where the MPC Margin Comes From

MPC differs from TUNED in three ways at once: it has an exact model of the plant, it sees a forecast, and it plans over a horizon. The arms of Table 3 form a ladder that separates them: THERM, TUNED, the D1 shield (an exact one-step model with no horizon), MPC at 0.25, 1, 2, 6, 24 and 48 h with a perfect forecast, MPC with a persistence forecast, MPC confined to the rule set, E2, and finally CLV and the bound. Fig. 4 draws five of its rungs against the clairvoyant bracket.

Against TUNED. With a perfect in-horizon forecast the 24 h MPC avoids 33.3, 32.0 and 29.8% of on-commands against TUNED's 22.8, 24.2 and 21.7%, and its cost saving exceeds TUNED's in every held-out window of every city, by Hodges–Lehmann estimates of +6.0, +4.2 and +4.6 points (family F2, 95.3% intervals $[+3.8, +7.5]$, $[+3.8, +5.3]$ and $[+2.0, +8.0]$). With a persistence forecast the difference is −0.1, +0.9 and −1.9 points, with intervals that include zero and pooled differences under 3 points: not resolved (F3). Forecasts between persistence and perfect were not tested. The persistence arms run 0.09, 0.07 and 0.09 K cooler than TUNED, in Riyadh and Taif at the 0.10 K fallback, which by the comfort-offset caveat penalises them; the back-off knob is tried only when no λ is admissible at 0.10 K, so it was not applied to these cells.

Horizon and model. The horizon sweep is descriptive (Table 3, with its back-off and fallback marks). Savings rise with the horizon up to 6 h,

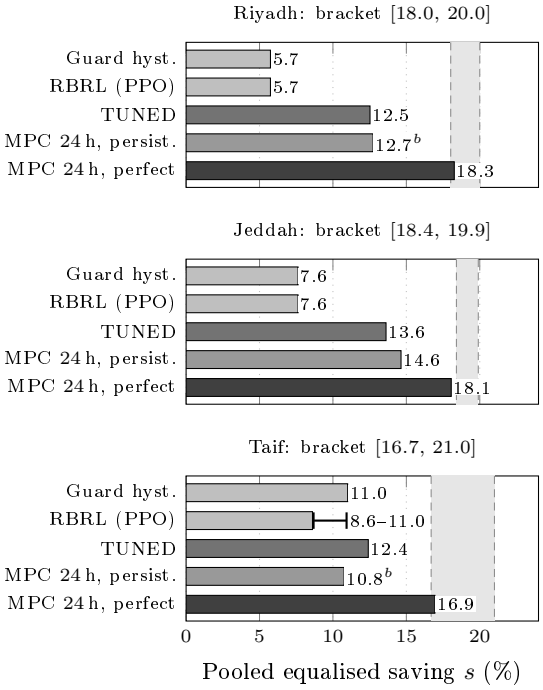


Figure 4: Savings ladder: pooled equalised saving s of five arms in each city, with the clairvoyant bracket [CLV raw, B^*] shaded and its ends dashed (values of Table 3; B^* under V_N, Section 8.1). Guard hyst.: the guard hysteresis; RBRL: rule-constrained PPO, the five seeds' range drawn as an interval where it exceeds the rounding. [b]fitted at the pooled 0.10 K fallback. The 24 h MPC's pooled margin over TUNED, 4.4 to 5.7 points with a perfect forecast, becomes −1.6 to 1.0 with a persistence forecast (not resolved); RBRL stays at the guard hysteresis' level in Riyadh and Jeddah, and in Taif its seeds reach down to 8.6% against the guard's 11.0%, a difference under 3 points that is not resolved.

and differences of less than 3 points beyond it are not resolved; the Taif 48 h cell, fitted at the fallback and 0.09 K warmer than THERM, exceeds the clairvoyant lower end. The 0.25 h MPC saves at most 2% and costs more than the baseline in Riyadh (−3.9%). The one-step D1 shield on a latch saves 14.0, 13.5 and 12.4%, within 1.4 points of TUNED in every city, while the 24 h MPC exceeds it by 4.3, 4.6 and 4.5 points: an exact one-step model reaches the tuned thermostat, and the margin beyond it needs the horizon and the forecast.

Forecast. At 6 h a persistence forecast costs 0.7 points in Riyadh and 1.9 in Taif, leaving 14.4 and 14.0%, 1.9 and 1.6 points above TUNED (not resolved); in Jeddah no multiplier or back-off is admissible at 6 h. At 24 h it costs 5.5, 3.4 and 6.2 points, partly at a cooler matched comfort (Table 3).

Thermal capacity. The single-node capacity is assumed (Remark 1). In Riyadh, refitted at each capacity, TUNED saves −2.8, 12.5, 16.0 and 15.8% at 2,400, 4,800, 7,200 and 9,600 kJ/K, and the 24 h MPC 12.2, 18.3, 19.8 and 16.2%, so its margin over TUNED is 15.0, 5.7, 3.8 and 0.4 points (Table 7); in Jeddah and Taif TUNED moves from −1.4 and −1.3% to 15.8 and 16.0% (16.7 and 17.7% at 7,200 kJ/K). The on-commands TUNED avoids rise monotonically with capacity in every city (Riyadh −4.0, 22.8, 34.6 and 39.8%). Its cost saving does not: the 7,200 and 9,600 kJ/K figures differ by less than 3 points in every city, and at 7,200 kJ/K TUNED sits 0.033, 0.040 and 0.121 K warmer than THERM against −0.019, +0.029 and −0.018 K at 9,600, so their order is not resolved. MPC's drop from 19.8 to 16.2% comes with a 0.044 K cooler matched mean and 2.5% more electricity at 9,600 kJ/K, while the on-commands it avoids barely move (42.4 and 40.7%). On the light plant tested (2,400 kJ/K) the commissioned thermostat does not beat a 1 °C deadband: each on-step overshoots by 0.75 K, TUNED's fitted deadband shrinks to 0.96 °C, and THERM's own fails (8) narrowly; THERM's self-score under Rule C was not computed there, so the small negative savings are not resolved from zero. On a heavy plant TUNED captures almost all of what MPC does. The ordering and its size therefore depend on the assumed capacity, and because the capacity acting over a cycle is probably below the reference (Remark 1), the reference results lean towards the end favourable to TUNED, and both arms' savings against THERM are probably overstated.

Robustness of the comfort device. The fitted multipliers of the 6 and 24 h perfect-forecast controllers lie at $\lambda/\lambda^* = 1.002$–$1.024$. In the following cells the fitted multiplier lies well above λ^*: $\lambda/\lambda^* = 1.226$ and 4.577 at 48 h in Riyadh and Jeddah, 13.095, 2.326 and 6.218 for the 24 h persistence controller, and 2.089 and 1.801 in rules at 24 h in Riyadh and Taif. Near λ^* the saving is sensitive to λ (Section 5.5). Runs at fixed $\lambda^* \cdot \{0.99, 1.01, 1.05\}$ are not comfort-matched and are given, with their $\Delta\tilde{T}$, in Supplementary Table S2 only; grid refinement is reported in Section 5.5.

Inside the rules. Confined to the training rule set (23.80/22.40 °C) the 24 h MPC saves 11.6, 14.1 and 10.0%, below relaxed bounds in those rules of 14.9, 15.6 and 16.2%. Confined to each city's *fitted*

guard thresholds, the rules the guard hysteresis itself uses, it saves 6.7, 4.4 and 10.5% against 5.7, 7.6 and 11.0% for the guard: it does not beat the guard by a resolved margin in any city, and in Jeddah it falls 3.3 points short at a 0.03 K warmer mean, in a cell fitted at the 0.10 K fallback with $\lambda/\lambda^* = 0.60$. Inside the wider training rules, whose on threshold is 0.14–0.20 K higher, the 24 h MPC exceeds the fitted guard by 5.9 and 6.4 points in Riyadh and Jeddah despite a 0.10 and 0.04 K cooler mean, which penalises it; this compares different rule sets. Replacing the guard's fixed margins by the D1 shield's one-step prediction lifts the latch to TUNED's level. Applied to the trained PPO policies in place of their rule layer, D1 gives 12.8–14.1 and 11.2–13.5% in Riyadh and Jeddah, the latch's level, but 8.0–12.0% in Taif, below the latch's 12.4%, and for three of the five seeds below their own rule-layer results (seed 42: 8.0 against 8.6%; differences under 1 point, not resolved). Inside the guard's own fitted thresholds, then, neither lookahead nor learning adds a resolved margin, and the one-step shield sets the level (Remark 5).

8.4. Multi-Zone Dwellings

Whether the single-zone ordering survives in a dwelling of several rooms is tested in Riyadh for four layouts: the single zone, a 1×2 row, a 2×2 grid and a 1×4 row, each with the glazed faces at the fixed azimuths of Table S-MZ and $\lambda_{ij} = 12.5$ W/K, with 35 W/K as a sensitivity. Every room has its own unit, so there is no shared capacity limit; rooms are coupled only through the shared walls. The arms are THERM, TUNED, the guard hysteresis and a multi-zone MPC, all under the protocol of Section 7 applied to the dwelling's mean occupied temperature, with each room's mean and the spread between rooms reported beside it. The multi-zone controller is a non-cooperative distributed MPC with one-step-delayed plan exchange [34] (Supplementary Section S7), with no optimality guarantee, no bound, because the joint dynamic programme is intractable (Section 9.4), and no guarantee of constraint satisfaction on the coupled plant: because each room plans against its neighbours' plans of the previous step, it records 0.0001 °C·h of occupied over-temperature in the 1×4 row at $\lambda_{ij} = 35$ W/K and at most 0.00003 °C·h of commanded under-temperature in the 2×2 grid.

Among the threshold rules, TUNED saves, on the dwelling mean, 12.5% in the single zone, 12.4% in the 1×2 row, 11.7% in the 2×2 grid and 13.6% in the 1×4 row (12.2, 12.9 and 13.3% at $\lambda_{ij} = 35$ W/K), and the guard hysteresis 5.7, 7.2, 8.3 and 7.7%. The multi-zone MPC (24 h, perfect forecast) saves 16.0, 15.9 and 15.7% (15.8, 16.1 and 16.1% at $\lambda_{ij} = 35$ W/K; the 1×2 cells at the 0.10 K fallback), at 0.004–0.072 K cooler than THERM, which by the comfort-offset caveat penalises it, avoiding 28–32% of the equalised baseline's on-commands with electricity within ±0.7%; on the 1×1 instance it reproduces the single-zone 18.3% exactly (Table S-MZ). Its pooled margin over TUNED at $\lambda_{ij} = 12.5$ W/K, 3.6, 4.2 and 2.1 points, is not tested for significance, and the last is below the 3-point resolution; its shortfall from the single-zone 18.3%, 2.2–2.6 points, is not resolved either. Every room has the same band and timetable, so the rooms' mean temperatures differ by only 0.03–0.08 K and little heat crosses a wall: the study tests whether the single-zone ordering transfers to a coupled dwelling, not what coordination is worth.

9. State-of-the-Art Comparison and Ablation Study

This section compares the figures with published ones, ablates the rule layer and the learned component, gathers the sensitivities, including the actuation step, and states what limits scalability.

9.1. State-of-the-Art Comparison

Published RL-based HVAC studies differ in control type, building type, baseline and climate; Table 5 records the basis of each figure.

Xue et al. [32], whose protocol is closest to this one (a rule-based comparator and a cost objective), report 6.7%. The figures here differ in three ways. Two lower them: the comparator is shifted to the arm's own mean temperature, and every figure is held out. One raises them: the saving is priced on on-commands at the 0.15 SAR reference, above the derived range, at which TUNED would save 3.8–11.0% in Riyadh. Electricity is about unchanged here, whereas the published figures are energy or cost savings. The larger published figures differ in scope, not degree, as the notes record.

9.2. Component Ablation and Sensitivity

Table 6 takes TUNED as the reference and changes one component at a time: the switch points are replaced by the rule layer's thresholds, a learner, an exact one-step model or a 24 h horizon is added,

Table 5: State-of-the-art comparison. The "Basis" column states what each saving is measured against.

Reference	Year	Method	Saving	Basis
[10]	2020	RL, whole commercial building	22%	vs. handcrafted rule
[23]	2021	DDPG, multi-zone residential	15%	vs. DQN
[9]	2021	SAC/TD3, continuous	>10%	vs. rule-based
[29]	2024	DRL, continuous VAV	37%[a]	vs. rule-based
[54]	2024	Online RL, white-box	65%[b]	vs. no control
[32]	2025	GA-MADDPG hybrid	6.7%	vs. rule-based
This work	2026	TUNED / MPC-24 h / bracket, Riyadh	12.5 / 18.3 / 18.0–20.0%	vs. thermostat at equal mean, held out; on-commands priced at 0.15 SAR, electricity ≈0

[a] Continuous VAV modulation in a simulated office: different hardware and building type.
[b] Relative to a no-HVAC baseline; saving vs. thermostat is substantially lower.

or the rule layer is removed. Ranked by contribution, the exact one-step model adds most (D1 on a latch exceeds the guard by 8.2 and 5.9 points in Riyadh and Jeddah and by 1.4 in Taif), the 24 h horizon with a perfect forecast adds 4.3–4.6 points over D1, the learner adds nothing in Riyadh and Jeddah and nothing resolved in Taif, and the rule layer is needed for admissibility in this configuration.

The rule layer. Without R1 and R2, PPO trained on the same reward with a linear comfort penalty yields an admissible ceiling for one of five seeds in Riyadh, which then saves −3.5%, and for none of the five in Jeddah or Taif. That ceiling, 23.50 °C, is the lower edge of the search grid, so the null for the other nine runs may partly reflect the grid's range. This shows that this configuration fails to optimise without the rules; it does not show that a learner needs them.

The learned component. Under the rule layer, PPO on the cost-only reward reproduces the guard hysteresis in Riyadh and Jeddah: four of five seeds in each reproduce the guard's pooled saving to 10^{-5} (5.745 and 7.624%) and the fifth differs by less than 0.004 points. In Taif one seed equals the guard (11.0%) and the other four, at fitted ceilings of 23.84 or 23.86 °C, save 1.0–2.4 points less, which is not resolved. Scored with the training cost at the training rules, three of those four seeds are within 0.05% of the hold policy on the six tuning weeks (the fourth 1.4%) and 0.5–2.5% above it on the held-out weeks, so the shortfall appears on weather not seen in training. On the tuning weeks the hold policy is itself 0.4, 1.4 and 2.3% above the in-rule optimum (E1, Section 5.4), which bounds what a longer run on the same weeks could gain; longer training and other hyperparameters were not tested. E2 is not selected by Rule A and not comfort-matched; it stays within 0.16 on-commands per day and 2.6% of cost of the hold policy at the training rules. Because its multiplier sits near λ^*, where the λ term is flat in comfort, and hold is within 0.4% of the reward's optimum in Riyadh, E2 is consistent with, not evidence for, a learning null; its savings are not tabulated. D1 in place of the trained policies' rule layer lifts them to the latch's level in Riyadh and Jeddah but not in Taif (Section 8.3); the level is set by the shield, not by the policies, which on their tuning weeks sit within 0.9% of the hold policy's training cost except the Taif seed at 1.4%.

The price of the guarantee. Compared with TUNED, which has no guard bands, the guard hysteresis costs 6.8 and 6.0 points pooled in Riyadh and Jeddah (Hodges–Lehmann 6.6 and 6.0), in every window (family F4); in Taif the difference, −1.4 points pooled, is not resolved, with the guard 0.05 K warmer than TUNED. On the 0.1 °C hardware grid the Riyadh difference vanishes, and under the per-window admission criterion the guard exceeds TUNED in Taif (Section 8.2). What the guards buy is a guarantee under the nominal plant parameters: nothing in TUNED's construction prevents an over-temperature or a cooling step below the band on unseen weather (it records the latter in Jeddah and Taif), whereas R1 and R2 exclude both by construction for the nominal plant. Because the cold margin is derived from the nominal capacity, and the capacity effective over a cycle is probably

Table 6: Ablation relative to the tuned thermostat: pooled equalised saving s (%) on the held-out windows, with one component removed or added at a time, and Δ, the difference from TUNED in points, computed from unrounded values. Values from Table 3 and this section; ranges are over the five PPO seeds. [b]fitted at the pooled 0.10 K fallback. Differences under 3 points are not resolved. E2 (comfort term and forecast; Riyadh, three seeds) is not comfort-matched and is not tabulated.

		s (%)			Δ vs TUNED (points)		
Configuration	Component removed or added	Riyadh	Jeddah	Taif	Riyadh	Jeddah	Taif
TUNED (reference)	two fitted switch points; no rules, model or learner	12.5	13.6	12.4	—	—	—
Guard hysteresis	switch points replaced by the rule layer's thresholds	5.7	7.6	11.0	−6.8	−6.0	−1.4
RBRL, 5 seeds	+ PPO inside the training rules	5.7	7.6	8.6–11.0	−6.8	−6.0	−3.8 to −1.4
PPO without rules	− rule layer (admissible seeds)	−3.5 (1/5)	none of 5	none of 5	−16.1	—	—
D1 shield on latch	rules' fixed margins replaced by an exact one-step model	14.0	13.5	12.4	+1.4	−0.1	0.0
D1 on PPO, 5 seeds	+ PPO under the one-step shield	12.8–14.1	11.2–13.5	8.0–12.0	+0.2 to +1.6	−2.4 to −0.1	−4.4 to −0.4
MPC 24 h, guard rules	+ model, horizon and perfect forecast, inside the guard's fitted rules	6.7	4.4[b]	10.5	−5.8	−9.3	−1.9
MPC 24 h, training rules	the same, inside the training rules	11.6[b]	14.1	10.0[b]	−0.9	+0.4	−2.4
MPC 24 h, persistence	+ model and horizon, persistence forecast, no rules	12.7[b]	14.6	10.8[b]	+0.2	+1.0	−1.6
MPC 24 h, perfect	+ model and horizon, perfect forecast, no rules	18.3	18.1	16.9	+5.7	+4.4	+4.5

lower (Remark 1), that guarantee does not survive the capacity uncertainty: at 2,400 kJ/K the per-step cooling excursion (0.75 K) exceeds ϵ_{cold} (Section 5.3).

Sensitivities. Table 7 gathers the economic and physical parameters. The tariff and the charge per on-command move the percentage through the share of the baseline's cost that on-commands carry, as Remark 4 predicts. The comparator's deadband moves the saving most, from 40.2% at 0.5 °C to 5.7% at 1.25 °C, because it sets how often the baseline cycles: the reported figure is a statement about a pair, controller and comparator. The adopted $\delta_{\text{db}} = 1\,^{\circ}\text{C}$ is assumed, not measured. It sits near the upper end of the interval $[0.375, 1.25]\,^{\circ}\text{C}$ admissible when the comparator is sampled at 15 min, a cap that a continuously sensing thermostat does not share, so an installed thermostat could have a wider deadband, against which TUNED saves less (5.7% against 1.25 °C). Measuring the installed stock's deadband is the field test most likely to confirm or overturn these results.

9.3. Actuation Step

Every figure above is tied to the 15-min actuation step. Table 8 and Fig. 5 repeat TUNED, the 24 h MPC and the bound at 5 and 30 min, each arm refitted by Rule A. At 5 min the tuned thermostat avoids 43.1, 41.9 and 43.0% of on-commands, against 22.8, 24.2 and 21.7% at 15 min, with electricity within 1% of the equalised baseline's; its cost saving doubles to about 25% and reaches 92% of the Riyadh bound. Its difference from the Riyadh 24 h MPC is not resolved, and that MPC ran on a coarser grid (801 nodes) at the fallback, 0.11 K cooler than TUNED. At 30 min TUNED has no resolved saving over THERM: it issues 2.2–2.8% more on-commands and saves −1.2 to −1.5%, while the 24 h MPC avoids 12.5, 21.0 and 20.4% and keeps 6.6–11.4%. There THERM's own deadband fails (8) narrowly and THERM leaks 0.025 occupied degree-hours in the first held-out Riyadh week, which favours THERM. The mechanism is the deadband of Section 8.2: the fitted deadband is 1.78–1.79 °C at 5 min and 0.94–0.99 °C, about THERM's own, at 30 min, as the per-step excur-

Table 7: Sensitivity of the pooled equalised saving, Riyadh, seven held-out windows; descriptive, no tests. (a) Tariff schedule at $c_{\mathrm{sw}} = 0.15$ SAR. (b) Charge per on-command at the tariff in force; the derived range is 0.025–0.114 SAR. (c) THERM's deadband at a fixed $[22, 24]$ °C band; admissibility by (8); the comparator is sampled at 15 min. (d) Thermal capacity, every arm refitted by Rule A. Every TUNED and MPC cell is fitted at pooled 0.03 K; B^* under V_N. On-command share: $c_{\mathrm{sw}} f_s / J$ of THERM. Bold: reference setting. At 2,400 kJ/K THERM's 1 °C deadband fails (8) narrowly. TUNED is refitted at every charge; its switch points stay at 23.71/22.36 °C for every $c_{\mathrm{sw}} > 0$ because the comfort limits bind, and move to 23.54/22.39 °C at $c_{\mathrm{sw}} = 0$.

Setting	THERM on-command share	TUNED	MPC-24 h	B^*
(a) Tariff (SAR/kWh)				
0.18 (in force)	55.0%	12.5	18.3	≤ 20.0
0.21 (2027)	51.2%	11.6	—	≤ 18.6
(b) Charge per on-command (SAR)				
0	0.0%	0.0	—	—
0.025	16.9%	3.8	5.7	≤ 6.3
0.05	29.0%	6.6	—	—
0.10	44.9%	10.2	—	—
0.114	48.2%	11.0	16.1	≤ 17.5
0.15	55.0%	12.5	18.3	≤ 20.0
0.20	62.0%	14.1	—	—
0.30	71.0%	16.2	—	—
0.60	83.0%	18.9	—	—
(c) THERM deadband (°C; THERM on-commands/day)				
0.50 (7.37)	—	40.2	—	—
0.75 (5.22)	—	23.4	—	—
1.00 (4.10)	—	12.5	18.3	—
1.25 (3.45)	—	5.7	—	—
1.50		inadmissible by (8)		
(d) Thermal capacity (kJ/K)				
2,400 (THERM 7.43/day)	—	−2.8	12.2	—
4,800 (4.10/day)	55.0%	12.5	18.3	≤ 20.0
7,200 (2.94/day)	—	16.0	19.8	≤ 20.0
9,600 (2.24/day)	—	15.8	16.2	—

sion allows. The step also changes the sampling of the comparator, a 1 °C hysteresis evaluated every Δt. The headline numbers therefore describe a 15-min supervisory step; they are not transferable to other control steps.

9.4. Scalability

The single-zone dynamic programme of MPC has 1,601 grid states and two actions per step. Training cost, decision time and embedded deployability are not reported: the only timing run predates the MPC implementation used here and carries no record of the code that produced it, so no decision time is claimed.

What can be stated is structural. A clairvoyant programme over N_z coupled zones with G grid points per zone carries $G^{N_z} 2^{N_z}$ states (the on-command charge makes each zone's cost depend on its previous action) and 2^{N_z} joint actions, so the backward pass touches $G^{N_z} 4^{N_z}$ transitions per interval: 4.1×10^7 at two zones and $G = 1{,}601$, 2.6×10^{11} at three, 8.0×10^{60} at sixteen. That is why the bracket of Section 8.1 is a single-zone result and why the multi-zone study has no bound. Multi-zone results for a learned policy under the

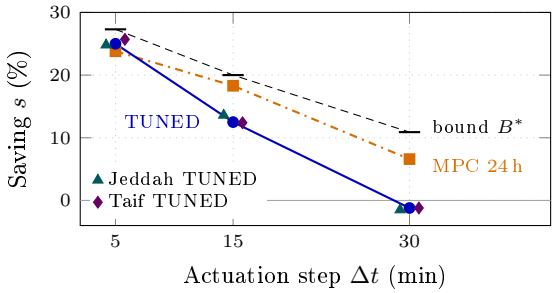


Figure 5: Pooled equalised saving against actuation step, as in Table 8. Lines: Riyadh TUNED, 24 h MPC (perfect forecast; at 5 min 801 nodes, 0.10 K fallback) and bound B^*; markers: Jeddah and Taif TUNED, offset horizontally. The gap between TUNED and MPC closes as the step shortens; at 30 min TUNED saves −1.2 to −1.5%.

Table 8: Saving s (%) against actuation step and capacity: TUNED, MPC-24 h (perfect forecast) and B^* (V_N, rounded up), every arm refitted by Rule A at pooled 0.03 K unless marked. Bold: headline step. —: not run; MPC and B^* at 5 min and at 7,200 kJ/K were run for Riyadh only, by design. At 30 min THERM's deadband fails (8) narrowly. [b]801 grid nodes (headline 1,601), pooled 0.10 K fallback, $\Delta\tilde{T} = -0.078$ K. [c]$\lambda/\lambda^* = 6.27$: for $\lambda \leq 0.21$ the controller holds the room about 0.2 K warmer than THERM on the tuning windows (pooled 23.12–23.14 against 22.91 °C, inadmissible), and at $\lambda \geq 0.29$ the mean jumps to 22.88 °C; the coarse step widens the sensitivity to λ of Section 8.3, the cheapest admissible λ is 1.51 (tuning cost 0.972 against 0.932 at the inadmissible 0.21), and the 6.6% is the protocol result, a conservative value. At 7,200 kJ/K TUNED is 0.033, 0.040 and 0.121 K warmer than THERM, so the comfort-offset caveat applies.

Δt, C_{th}	Arm	Riyadh	Jeddah	Taif
5 min, 4,800	TUNED	25.0	24.8	25.7
	MPC-24 h	23.8[b]	—	—
	B^*	$\leq$ 27.3	—	—
15 min, 4,800	TUNED	12.5	13.6	12.4
	MPC-24 h	18.3	18.1	16.9
	B^*	$\leq$ 20.0	$\leq$ 19.9	$\leq$ 21.0
30 min, 4,800	TUNED	−1.2	−1.5	−1.2
	MPC-24 h	6.6[c]	11.4	11.3
	B^*	$\leq$ 10.9	$\leq$ 13.9	$\leq$ 14.7
15 min, 7,200	TUNED	16.0	16.7	17.7
	MPC-24 h	19.8	—	—
	B^*	$\leq$ 20.0	—	—

same tariff, with a larger unit, a $[23, 25]$ °C band and Jeddah profiles, are in [12] and are not directly comparable.

10. Discussion

This section sets out how far the results can be trusted (Section 10.1) and how far they extend (Section 10.2), the factors that govern the saving (Section 10.3) and the limitations (Section 10.4).

10.1. Verification and Validity

Verification. The RC step (3) is solved exactly, so no result carries integration error (Section 7). Before any table is produced, a gate of 188 numerical invariants in nine classes (physical sanity, bound consistency, accounting identities, parameter provenance, monotonicity, decomposition closure, feasibility, the deadband actuation condition, and the protocol checks on every results file) exits non-zero on any failure; all pass, although the feasibility class retains some checks written for the band of an earlier version of the model. On the 1×1 instance the multi-zone simulator reproduces the single-zone results exactly (Section 8.4), and every results file records the code version that produced it.

Internal validity. Every parameter is selected on the six tuning windows and every saving is measured once on the seven held-out windows (Section 7); each arm is compared with a thermostat shifted to its own mean temperature (Rule C), and every arm satisfies $s_a \leq B_a$, the clairvoyant bound at that mean under V_N, in all 89 rows checked (Section 8.1).

Model validity. The envelope is parameterised from the SBC 2018 prescriptive limits (Section 6.2) and the plant data are assumed (Table 2). The first-order RC form belongs to the family used for control-oriented residential models [38, 39], but Bacher and Madsen [42] rejected a single state for the building they measured (Section 10.4). No metered data exist for a dwelling of this description, so the model is an uncalibrated reference: the ASHRAE Guideline 14 fit criteria cannot be computed, no calibration claim is made, and no co-simulation or hardware-in-the-loop test was run. The results are model-based comparisons under a stated plant, and absolute figures are not predictions for a real dwelling. Relative comparisons on one model are not immune to a shared bias either: the margin of MPC over TUNED moves with the assumed capacity, and TUNED has no resolved saving at half the reference capacity (Section 8.3) or at a 30-min step (Section 9.3), so every comparison is to be read with those sweeps.

10.2. Generalisability

What holds. TUNED saves 12.5, 13.6 and 12.4% in hot-dry Riyadh, hot-humid Jeddah and highland Taif. The differences reflect load, diurnal swing and the baseline's cycling rate more than any algorithm: Riyadh's 12.7 K swing lets the zone free-float further than Jeddah's 8.4 K, and Taif's load varies from week to week, which a fixed pair of switch points cannot follow and MPC with a perfect forecast can (with persistence it saves 10.8%). On three actual years per city (Supplementary Section S5; ERA5 is not the station record behind TMYx, so the spread mixes year-to-year variation with dataset differences) TUNED saves 10.5–13.4% with the typical-year thresholds and 11.5–14.8% retuned, though in Taif's two milder years it free-floats below the band for 2.6–6.9 °C·h. With the Riyadh façade turned to each cardinal direction

it saves 12.2–13.3%, and in two- and four-room dwellings 11.7–13.6% (Section 8.4).

What moves. TUNED's Riyadh saving runs from −2.8 to 16.0% across 2,400–9,600 kJ/K (Section 8.3); across actuation steps of 5–30 min from about 25% to no resolved saving (Section 9.3); on a 0.1 °C setpoint grid it falls to 5.7–8.4%; and at the 2027 rates it is 11.6–12.7% for the same on-commands avoided (Section 8.2).

Where the findings stop. They are established for single-speed on/off units with a constant COP, an account that stays inside the first tariff block (Remark 3), so that the price is flat in practice, fixed occupancy, no internal gains or latent load, and no field data; Section 10.4 gives each restriction.

10.3. What Governs the Saving

In Riyadh the comparator's deadband moves the saving by 35 points across its admissible range, the controller class by 13 (guard 5.7, TUNED 12.5, 24 h MPC 18.3%), the charge per on-command by 7 across the derived range and the 2027 repricing by 0.9 (Table 7). Only the controller class is algorithmic. The actuation step moves the saving further (Section 9.3).

The band acts mainly through the baseline. Narrowing it gives an optimising controller *less* room, but THERM, which switches 0.5 °C inside each edge, cycles more often on a narrower band, so the saving rises; the band was not swept under the protocol, so no figure is given. The narrower band is appropriate for this stock (Section 6.2), but any percentage reported for binary scheduling must be read with the band it was measured on.

What does move electricity is the comfort target. Along THERM's own equalisation curve on the held-out windows, inside the 24 °C ceiling, raising THERM's switching thresholds by 0.1 K (a 0.10–0.12 K warmer occupied mean) lowers its electricity by about 0.9, 1.4 and 2.3% in Riyadh, Jeddah and Taif, at the price of a warmer room; at matched comfort no switching policy changes it.

The tariff acts through the denominator. At matched mean temperature scheduling changes only the number of on-commands, each worth c_{sw}, so the on-commands TUNED avoids are the same at the 2018 and 2027 rates while the percentage falls as the bill grows (Section 8.2). A percentage quoted at one tariff is not comparable with one quoted at another. With a constant marginal price (Remark 3) and no time-of-use element, the store the dwelling holds (0.89 kWh electrical equivalent across the band) offers nothing to arbitrage; a time-varying price would change that.

A controller could lower the electricity bill, rather than the on-command charge, only by changing the electricity drawn. In the single zone, MPC and the clairvoyant bound already choose the whole temperature path inside the band at every step; in the coupled dwellings the distributed MPC does so room by room. None of them pre-cools (Remark 2): with a single-speed unit, a constant COP and a flat price, cooling earlier only adds the energy the store leaks. Electricity at matched comfort becomes policy-dependent only when the cost of *delivered* cooling varies over the day, through a COP that rises at night (the 15–20% of Section 5.3), a time-of-use price, or a multi-speed or inverter unit whose part-load efficiency differs from full load. None of these is modelled here (Section 10.4); a controller that chooses speed and timing jointly across coupled rooms under those conditions, with setback in every room, is the natural next study.

Implications for Saudi Vision 2030 and Green Riyadh.. The results separate the levers open to the national programmes. At matched comfort the controllers studied here change on-commands, not electricity, so in this model they contribute to emission cuts only through an assumed start-up loss per on-command (Section 10.4). Electricity, and with it emissions at a given generation mix, is governed by the comfort target and the outdoor air. Two consequences follow. First, setpoint and urban-cooling measures are where household electricity savings lie: each 0.1 K warmer comfort level lowers THERM's cooling electricity by 0.9–2.3%, and outdoor air 1.5 and 2 K cooler, the Green Riyadh aim, lowers it on the held-out weeks by 10.5 and 13.7%, a quantified energy co-benefit of urban greening for each dwelling. This is a first-order upper estimate: the shift is uniform over all hours of an airport weather year, the plant and thresholds are unchanged, the model has no internal gains, and a constant COP omits the efficiency gain in cooler air; TUNED's raw saving there (12.8–13.6%, thresholds not refitted, 0.03–0.05 K warmer than THERM) leaves open whether a cooler city and better switching add. Second, retuning existing thermostats needs no new equipment beyond a 15-min control layer and avoids about a fifth of on-commands with no loss of comfort at the modelled resolution (fewer on a 0.1 °C setpoint grid), which bears on equip-

ment life and maintenance rather than on the bill. The clairvoyant bracket tells programme designers how much any switching controller could add, a saving of at most about 17–21%, so effort on control can be weighed against setpoint, efficiency and urban measures, and the open code and data allow the comparison to be repeated for other Saudi cities and dwellings.

10.4. Limitations

Supplementary Table S6 gives the direction in which each idealisation biases the reported percentage.

- *The saving is specific to six modelling choices.* The comparator's deadband, the actuation interval, the charge per on-command, the tariff, the assumed thermal capacity and the thermostat's setpoint resolution each move the headline by more than the resolution of the comparisons (Tables 7 and 8; Supplementary Tables S1 and S2), and the comfort band, which was not swept, acts on it through the baseline (Section 10.3); any figure quoted from this paper should be quoted with all of them. The deadband sweep is capped at 1.25 °C by sampling the comparator at 15 min, and the comparator's 1 °C deadband is assumed, not measured.
- *Tied to the* 15*-min step* (Section 9.3). The comparator is a 1 °C hysteresis evaluated at the actuation step; a thermostat that senses continuously is not represented, and the saving against it is not established.
- *On-commands, not compressor starts.* The model issues supervisory on-commands to a lumped node; whether each is a compressor start depends on the unit's own thermostat, and the model's on-command rate is an artefact of that interface, not a field prediction.
- *Single-node thermal network and assumed capacity.* Air and fabric share one capacitance. Bacher and Madsen [42], whose identification framework is followed, rejected the single-state model for the building they measured, because its residuals were not white, and selected a four-state grey-box model; a plant with a separate air node would cycle faster than this one. The capacity is assumed, and both results depend strongly on it, probably in TUNED's favour at the reference value (Remark 1, Section 8.3).
- *Thermostat resolution.* TUNED's switch points are fitted on a 0.01 °C grid; restricted to the 0.1 °C steps of a real thermostat it saves much less (Section 8.2), so the commissioning gain depends on hardware that can resolve the thresholds. The guard and RBRL ceilings are fitted on the same fine grid. Comfort matching to hundredths of a kelvin and every switch point assume noise-free sensing for every arm, whereas typical thermistors err by a few tenths of a kelvin.
- *Perfect model and forecast.* MPC plans on the plant's own model with the exact state and, in its perfect mode, the true in-horizon weather; the persistence mode removes the second idealisation but not the first, and model mismatch is untested. Its terminal rollout uses persistence weather and the TUNED rule, so no closed-loop dominance over TUNED is guaranteed. The PPO agent's one-hour outdoor value is also exact.
- *Window-end terminal effect.* The terminal store is priced at re-cooling energy without an on-command charge, so an oracle that knows the window end can save up to one on-command per window, and under that terminal CLV ends its windows warmer than THERM. The oracles therefore use V_N, which charges one on-command per kelvin left above the start temperature; that rate is a convention.
- *Plant data.* The 2 kW capacity and the COP of 2.985 are assumed, not taken from a rated unit. The COP lies below the SASO 2663:2018 minimum for split units at the T1 rating condition (about 3.46) though above its T3 minimum (about 2.43) [55]. A higher COP lowers the energy term and so raises the on-command share of cost and the percentage saving. Capacity and COP are constant, whereas a split unit loses capacity and efficiency at high outdoor temperatures; lower capacity acts in the hottest hours like a higher C_{th} in the actuation ratio. An outdoor-dependent COP (capacity held constant) is implemented in the repository but not reported; the effect of either derating on the comparison is not established.
- *No per-start energy and no internal gains.* An on-command costs money but no energy, so it cannot register as an energy saving; a transient start-up loss of 10–25 Wh would turn TUNED's energy difference into a saving of 0.33–0.80% in

Riyadh, 0.37–0.90% in Jeddah and 0.34–0.83% in Taif. Occupancy, lighting and appliance gains, of order 150–300 W in an occupied room and so comparable with the modelled envelope load, are omitted, as is solar gain on opaque walls (an intermediate-floor room is modelled). They shorten free-float and lengthen on-phases, which is the mechanism the capacity and step sweeps act through; their effect on the headline savings is not established, and they flatter any setback's recovery.

- *Comfort offsets.* Rule C does not cancel a difference in comfort between arms: along the guard family a 0.01 K warmer mean is worth about 1.3 points, so the 3-point resolution rule corresponds to about 0.02 K, below Rule A's 0.03 K tolerance. Arms whose $\Delta\tilde{T}$ differs by more than about 0.02 K, including those fitted at the 0.10 K fallback, are compared with that caveat. The admission rule itself matters: refitting under a per-window 0.03 K criterion moves TUNED by up to 8.4 points and reverses F4 in Taif (Supplementary Table S2); the MPC arms were not refitted under it, so the robustness of F2 and F3 to the admission rule is untested. Where the λ device cannot reach the matched comfort level, MPC is matched by backing off its occupied ceiling, and a negative saving there is a finding about the device, not a failure of the protocol.
- *Latent load.* The model is sensible-only. The latent axis was not tested at adequate fidelity, so for Jeddah, whose infiltration load is majority latent, this paper reports cost at matched dry-bulb comfort only, and not compliance with the ASHRAE 55 humidity criterion.
- *Scope of the learner.* PPO is tested only inside the rule layer and, in E2 (Riyadh, three seeds), with one comfort-aware reward; learners acting outside R1–R2 are untested. PPO is trained on six weekly weather sequences with one untuned hyperparameter set, so its held-out behaviour tests generalisation from those six weeks; longer training was not run, and on those weeks it could gain at most E1's margin (Section 9.2). R1 and R2 impose no minimum dwell, so the complete-cycle structure is not enforced; a minimum-dwell rule set is required work.
- *One year, one zone type, simulation only.* The held-out weeks are interleaved within one TMYx year; the headline is a single zone with one exposed face; the multi-zone study is one city, independent units, the same band in every room and no bound, and a shared capacity-limited plant is not studied. Time-of-use or demand tariffs, minimum on- or off-times, a temperature-dependent COP and multi-speed or inverter units are not evaluated, and the conclusions do not extend to them; setback is evaluated in the single zone only. No result is reported on a community benchmark such as BOPTEST [56], and co-simulation and hardware-in-the-loop tests are required before any deployment.

11. Conclusions

Binary on/off cooling of Saudi residential split units was formulated as a constrained optimisation with hard comfort constraints, a charge per on-command and an energy cost, and every controller was selected and evaluated under one protocol in Riyadh, Jeddah and Taif. Four findings emerge.

First, commissioning a two-parameter thermostat widens its deadband from the assumed 1 °C to 1.32–1.36 °C and avoids 22–24% of on-commands pooled, and fewer on-commands in every held-out window, with electricity within ±0.5%; priced at the derived 0.025–0.114 SAR per on-command this is worth 2–19 SAR per room per cooling season. The gain is a statement about the comparator as much as about tuning: against a 1.25 °C deadband the tuned thermostat saves 5.7% in Riyadh instead of 12.5%, on a 0.1 °C setpoint grid it avoids 10.5–14.7% of on-commands, and at its own mean temperature it attains 62–65% of the provable bound.

Second, the comparison depends strongly on the plant and the step. A 24 h MPC avoids 30–33% of on-commands and adds 4–6 points of cost saving with a perfect in-horizon forecast; with a persistence forecast, the only imperfect one tested, its margin is not resolved. In Riyadh the tuned thermostat has no resolved saving at half the reference capacity and saves 15.8% at twice it, while MPC's margin falls from 15.0 to 0.4 points; because the capacity acting over a cycle is probably below the reference, the reference results probably favour the tuned thermostat. At a 5-min step the tuned thermostat saves about 25% and at 30 min it has no resolved saving, so every figure here holds for a 15-min step only.

Third, PPO trained under hard rules reproduces the rules' hold policy in Riyadh and Jeddah and falls below it in Taif by a margin that is not

resolved, and a comfort-aware variant, tested in Riyadh only, stays close to hold. Inside the guard's own fitted thresholds neither lookahead nor learning adds a resolved margin; an exact one-step shield in place of the rules' fixed margins reaches the tuned thermostat's level on a latch, and inside the wider training rules a 24 h MPC gains about 6 points over the guard in Riyadh and Jeddah. The guarantee the guards give, which holds for the nominal plant, costs 6.8 and 6.0 points in Riyadh and Jeddah.

Fourth, at matched mean temperature every arm uses about the same electricity, by an identity of the model, so the cost difference is a charge on avoided on-commands, assumed to stand for wear. Electricity follows the comfort target, each 0.1 K warmer mean lowering THERM's by 0.9–2.3%, and the outdoor air: in a first-order sensitivity with no internal gains, a city 1.5–2 K cooler (the Green Riyadh aim) lowers one room's cooling electricity by 10.5–13.7% in Riyadh. For Saudi Vision 2030 and Green Riyadh, the work indicates that household electricity savings come from the comfort target, efficient equipment and a cooler city, while tuned switching reduces on-commands, and with them wear, at no cost in comfort. The money saved depends on the six modelling choices of Section 10.4 and on the comfort band, which was not swept.

Occupancy setback and multi-room dwellings were also examined. Setback raises the on-commands the tuned rule avoids to 29–35% and its cost saving to 17–20%. In Riyadh dwellings of two and four rooms with independent units the tuned thermostat saves 11.7–13.6%, against 12.5% in the single zone, and a non-cooperative distributed MPC saves 15.7–16.1%, against 18.3%; neither difference is resolved.

For this model instance three recommendations follow. (i) Judge any learner or planner against a thermostat whose deadband has been widened to what the band and the actuation step admit, not against the factory deadband. In practice such commissioning assumes a supervisory 15-min control layer and thresholds resolved to 0.01 °C (on a 0.1 °C grid the saving is 5.7–8.4%), and no commissioning or hardware cost is set here against the 2–19 SAR per season. (ii) Check both sides of the actuation condition (8) and derive guard bands from the per-interval excursion in each direction before trusting any reported saving. (iii) Report a saving with the band, the comparator's deadband, the charge per on-command and the on-commands avoided, and do not present it as a bill saving. Every figure is simulated, and field validation, with the installed stock's deadband measured, is required before any deployment claim.

12. CRediT Author Statement

Hamzah Faraj: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization.

13. Declaration of Competing Interest

The author declares no competing interests.

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15. Ethics Declaration

This research does not involve human participants, animal subjects or sensitive data. Ethics approval and consent to participate are not applicable; consent for publication is not applicable.

16. Data and Code Availability

What is released. The code is available at the project repository on GitHub, archived on Zenodo at doi:10.5281/zenodo.24199676 (release v4.0-round4); the trained PPO models (29 MB) are not in the archive, are available from the author on request, and can be retrained from the code with the stated seeds. The repository comprises the thermal model, every controller and oracle, the protocol module that applies Rules A–C to every arm, the statistics, and the scripts that generate every table from the results files, each of which records the code version that produced it. The invariant gate of Section 10.1 runs before any table is produced. The time-step and capacity comfort records of THERM in Supplementary Table S1 come from a separate THERM run on the first held-out Riyadh week. Code paths that the paper does not report (an outdoor-dependent COP, a behaviour-cloning variant, a shared-plant multi-zone arbitration) are in the repository and marked as unreported.

What is not available. An exploratory pipeline that preceded this study is archived in the repository (`archive/withdrawn-pipeline`) for transparency only; it does not reproduce, and is not the source of, any number or figure reported here.

Weather data. All three records are TMYx files from climate.onebuilding.org: Riyadh (WMO 404380, 2009–2022), Jeddah (WMO 410240, 2009–2023) and Taif (WMO 410360, 2009–2023) [57]. The source reserves its rights, so the EPW files are not redistributed; `fetch_weather.py` names each station file and verifies the downloaded copy by SHA-256. The actual-year series of the inter-annual study are ERA5 reanalysis [58] retrieved through the Open-Meteo archive API, redistributed in the repository under CC BY 4.0 with attribution (weather data by Open-Meteo.com; contains modified Copernicus Climate Change Service information).

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